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Tremblay et al.

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(45) **Date of Patent:** **Aug. 26, 2025**

(54) **QUICK LOCK RELEASE COUPLING**

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F16L 37/32 (2006.01)
F16L 37/23 (2006.01)
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(52) **U.S. Cl.**
CPC **F16L 37/32** (2013.01); **F16L 37/23** (2013.01); **F16L 37/24** (2013.01); **F16L 37/34** (2013.01); **F16L 37/40** (2013.01); **F16L 37/53** (2013.01)

(58) **Field of Classification Search**

CPC . F16L 37/40; F16L 37/34; F16L 37/32; F16L 37/53; F16L 27/023; F16L 27/08; F16L 37/23

See application file for complete search history.

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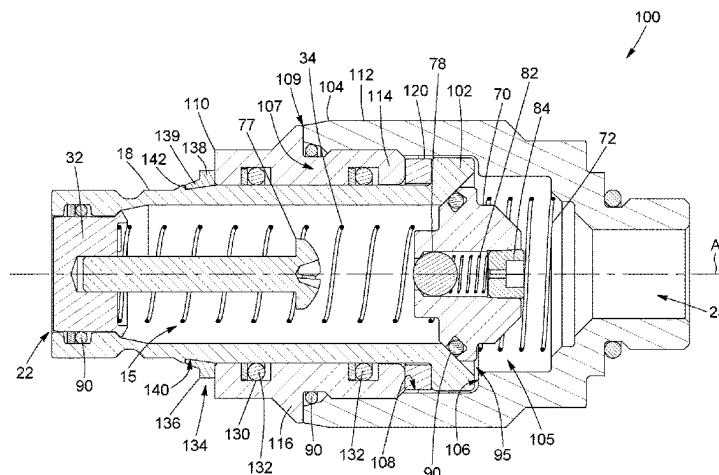
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(57) **ABSTRACT**

A quick release coupling for providing a connection between a pair of hydraulic lines is provided. The coupling includes a male component has a fluid passage therethrough and includes a shank portion adapted to engage a first hydraulic line. The shank portion has a first port for establishing fluid communication between the first hydraulic line and the fluid passage. The shank portion also has a restraining mechanism adapted to engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line. The coupling also includes a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage. The coupling adapter is adapted to engage a second hydraulic line and has a second port for establishing fluid communi-

(Continued)



cation between the second hydraulic line and the fluid channel.

18 Claims, 25 Drawing Sheets

(51) Int. Cl.

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F16L 37/34 (2006.01)
F16L 37/40 (2006.01)
F16L 37/53 (2006.01)

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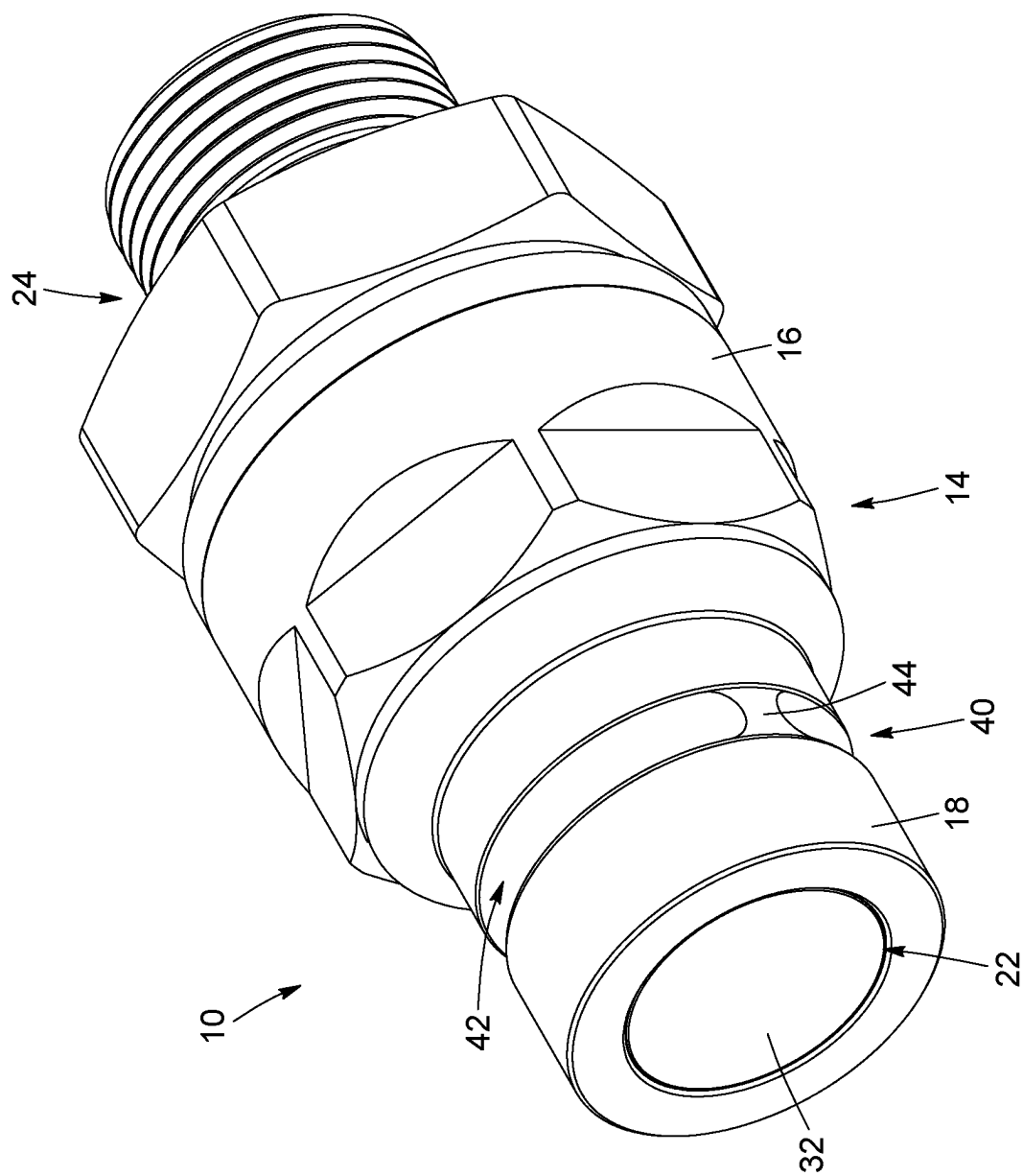


FIG. 1

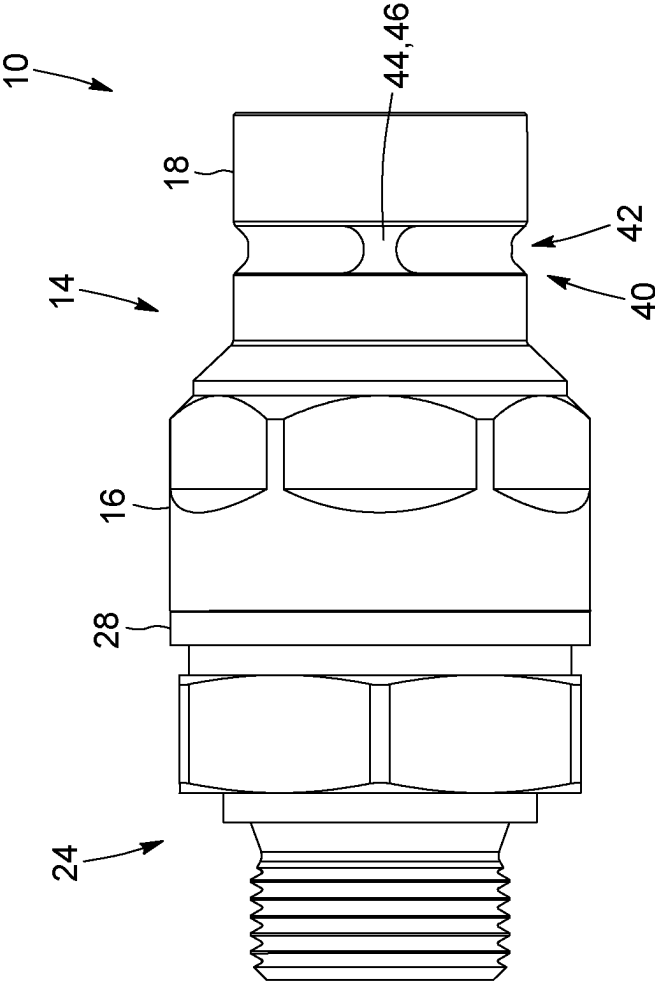


FIG. 2

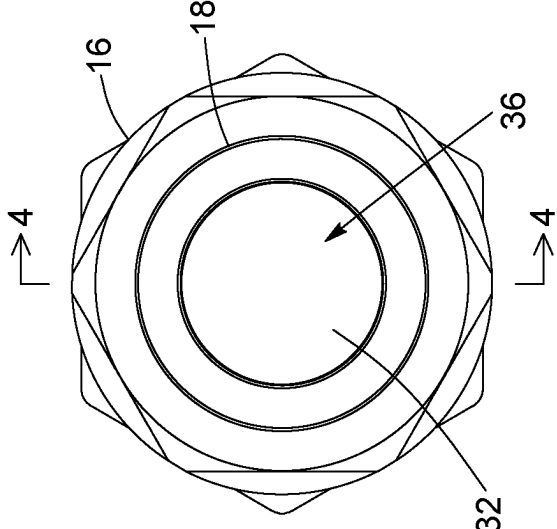


FIG. 3

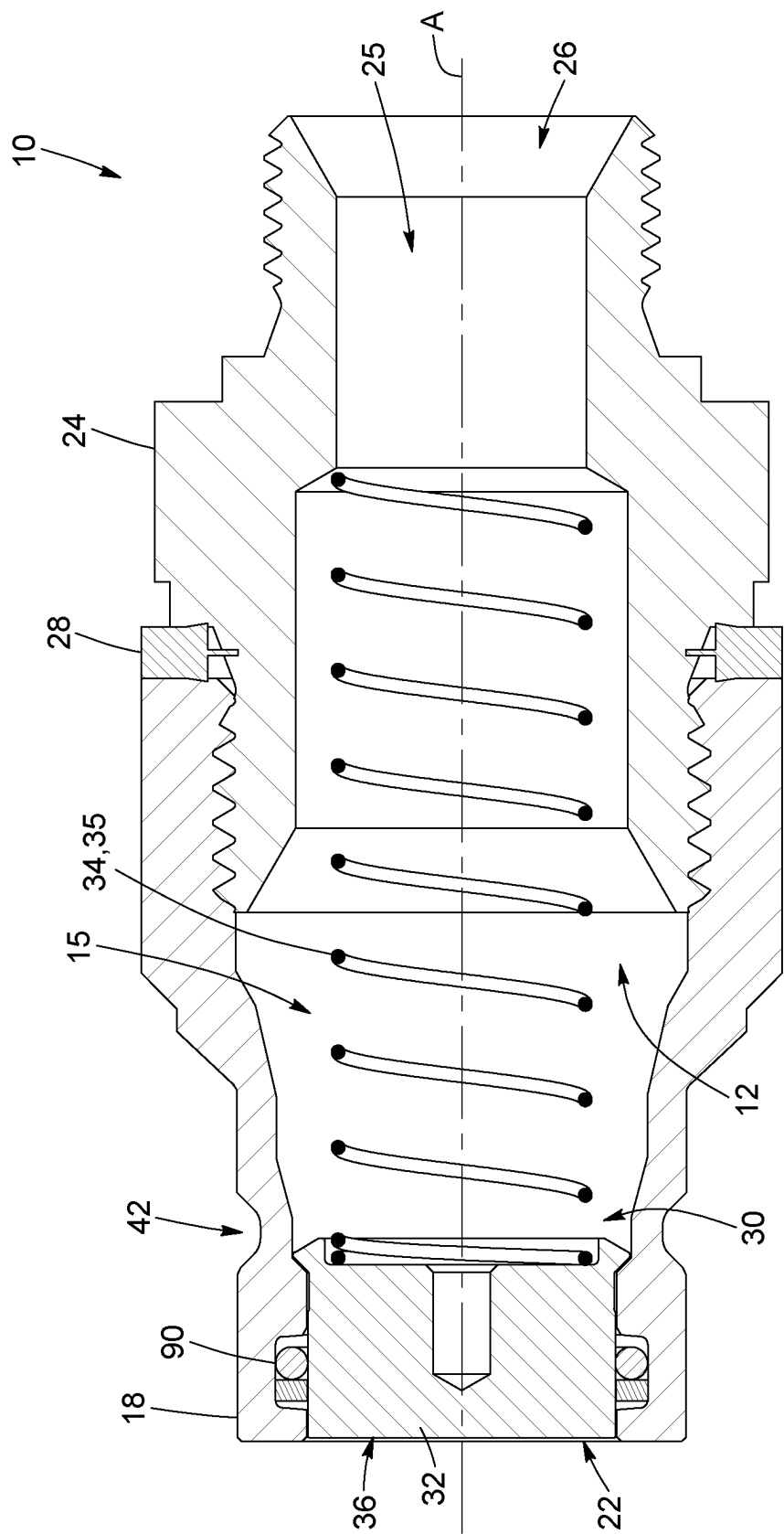


FIG. 4

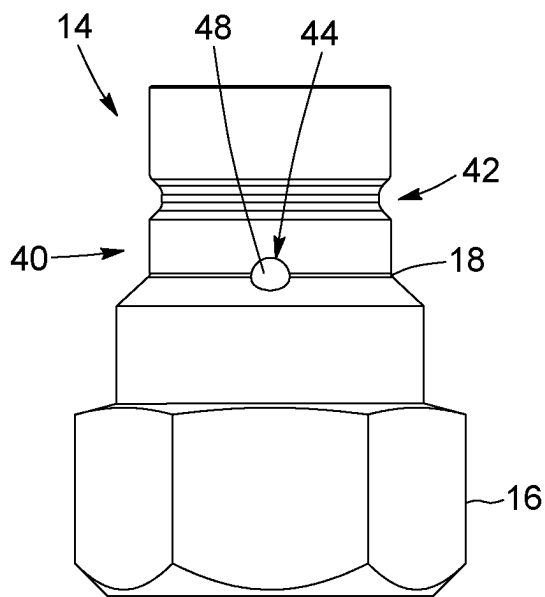


FIG. 5

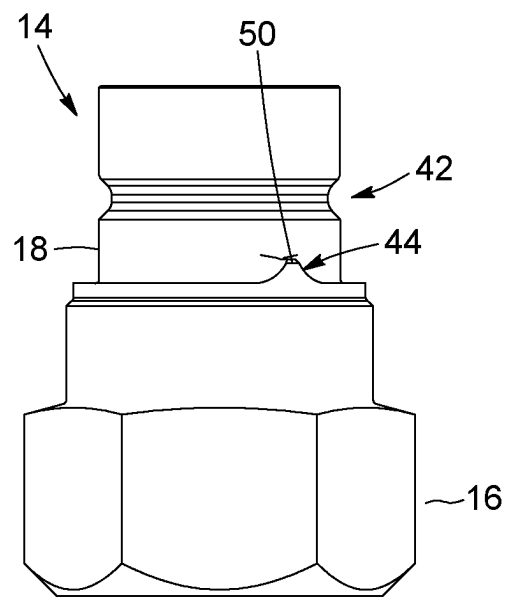


FIG. 6

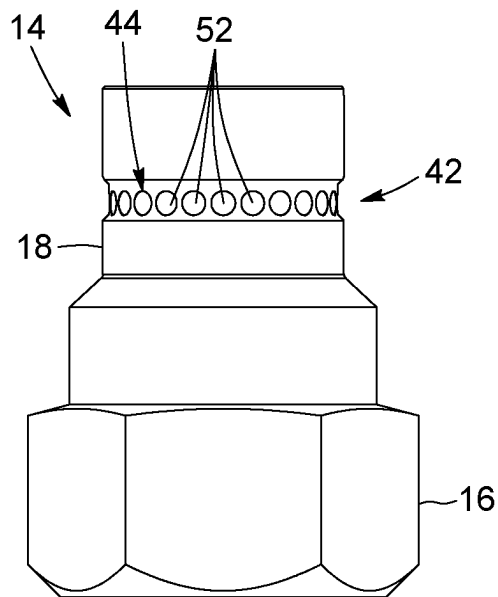


FIG. 7

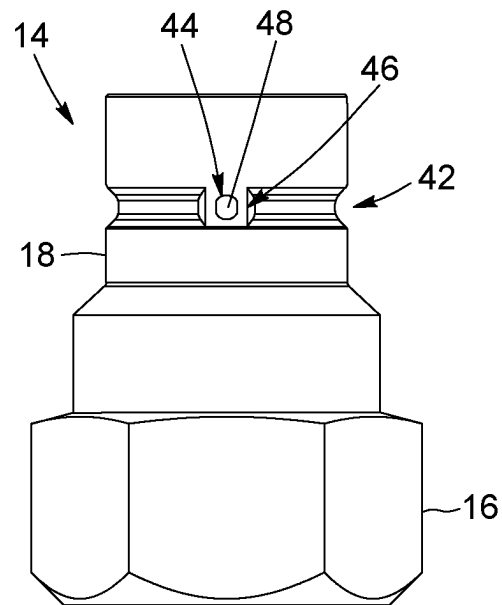


FIG. 8

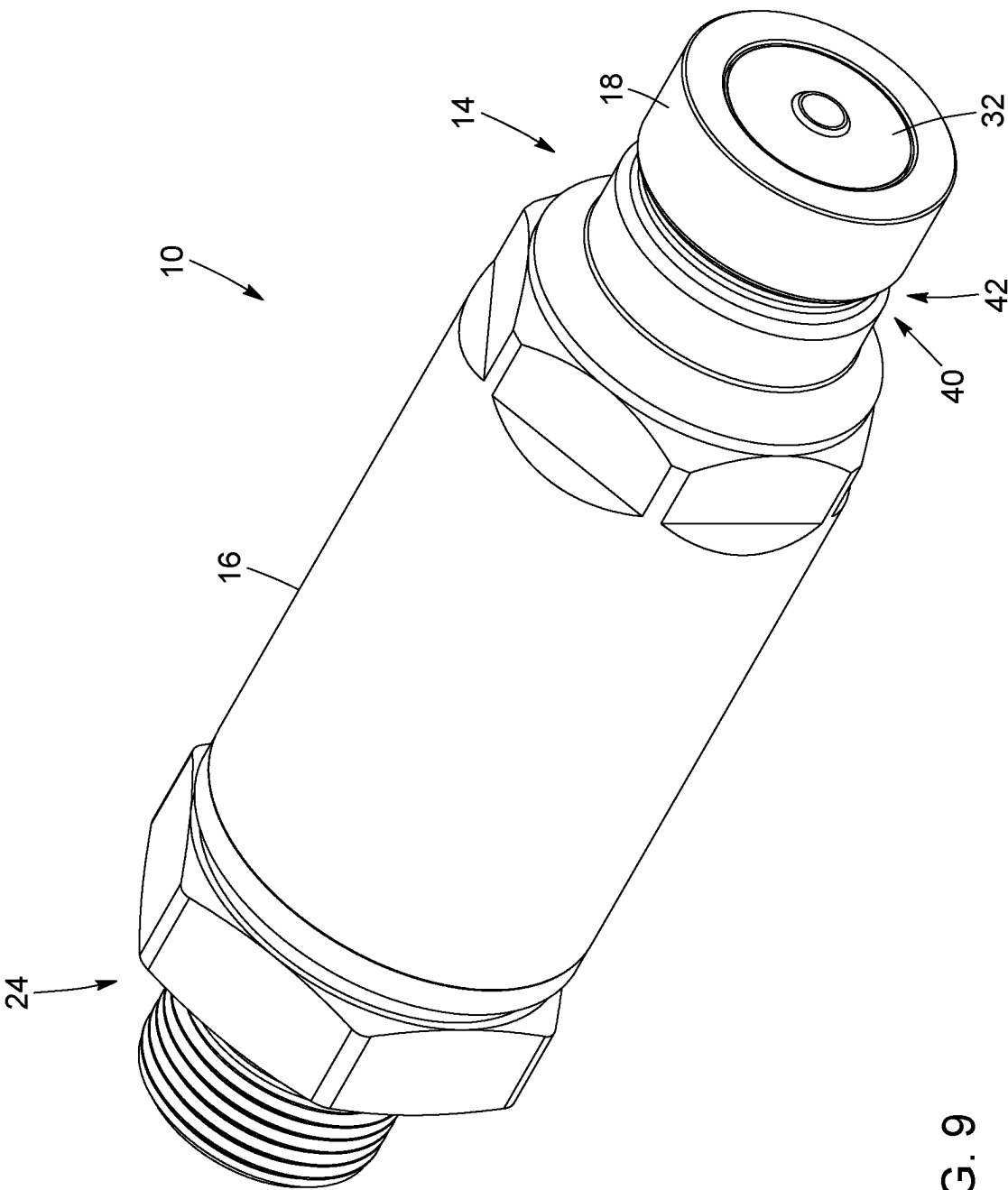


FIG. 9

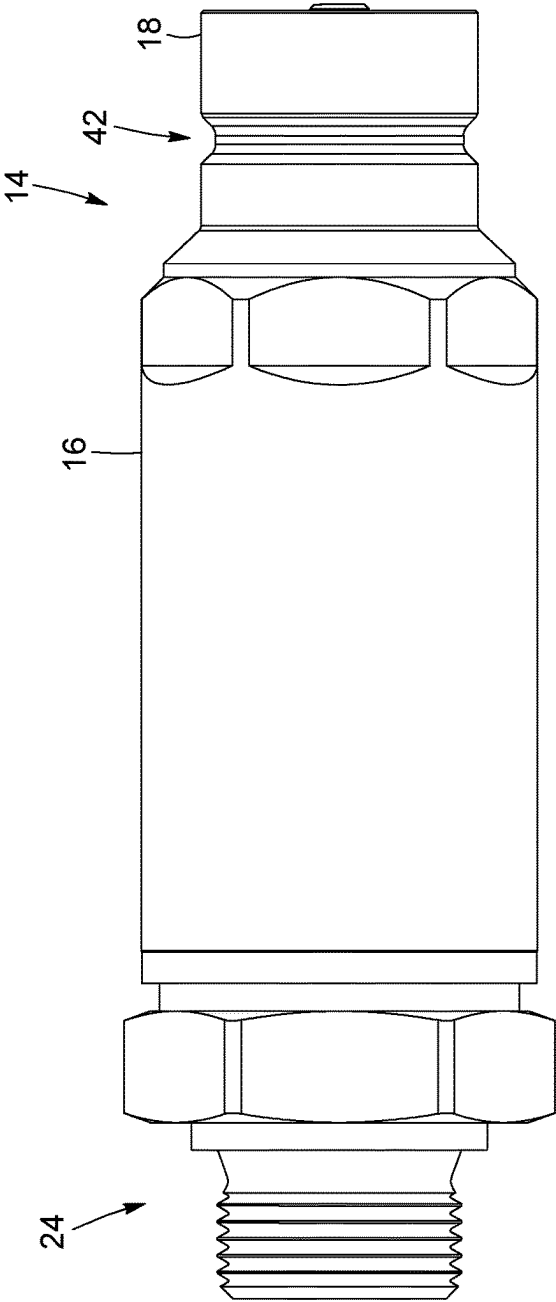


FIG. 10

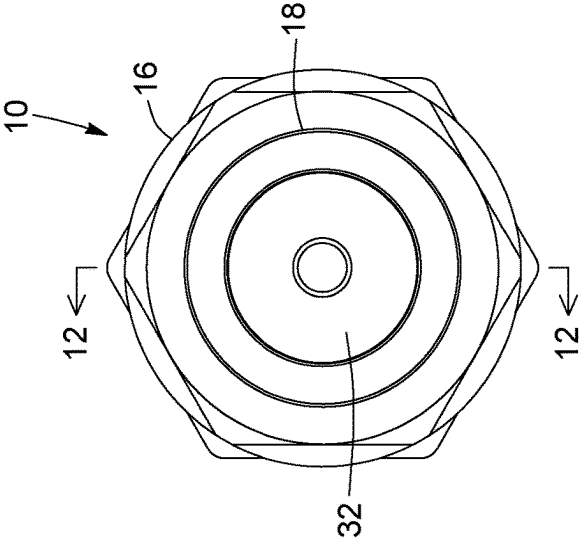


FIG. 11

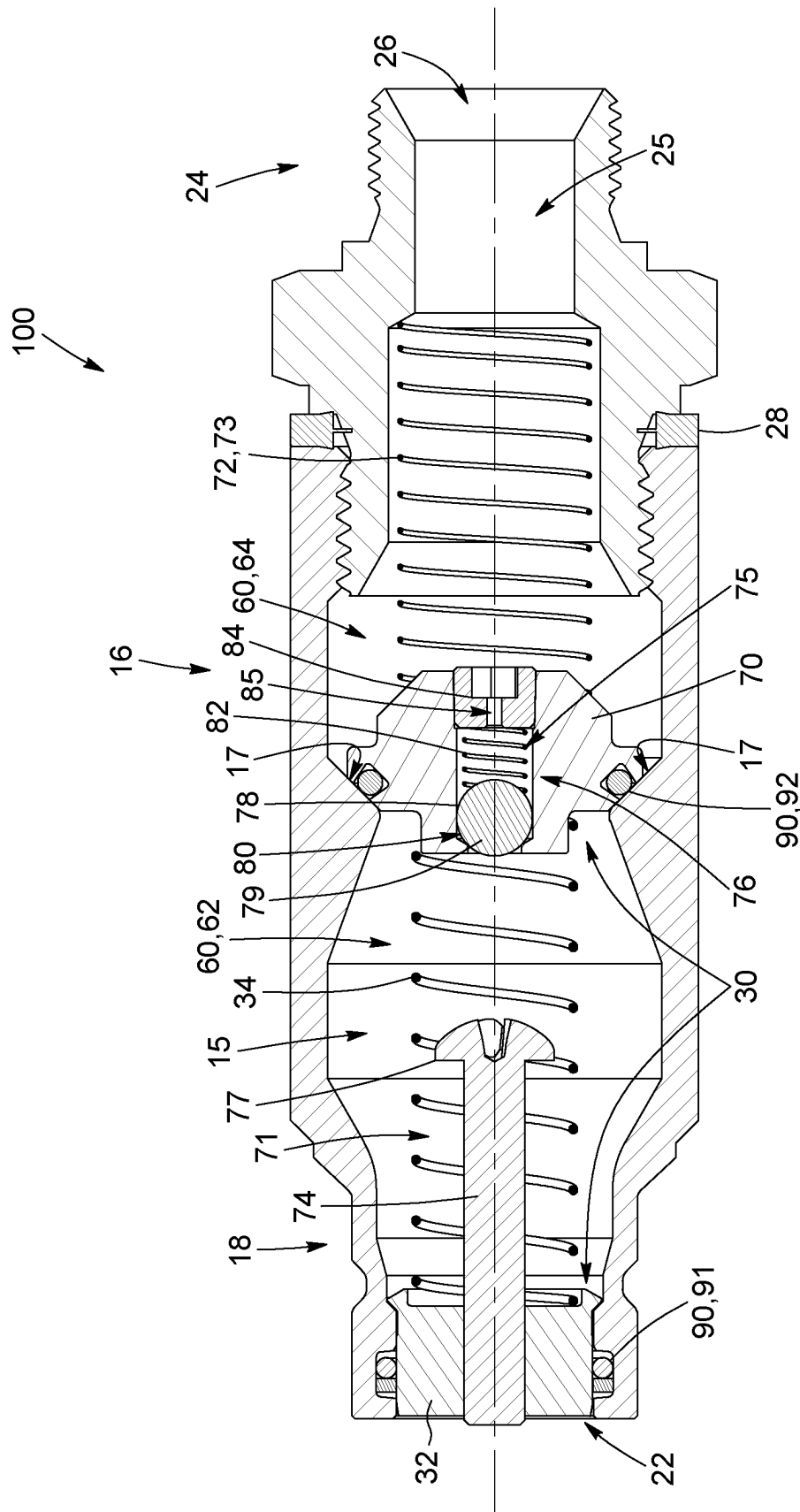


FIG. 12

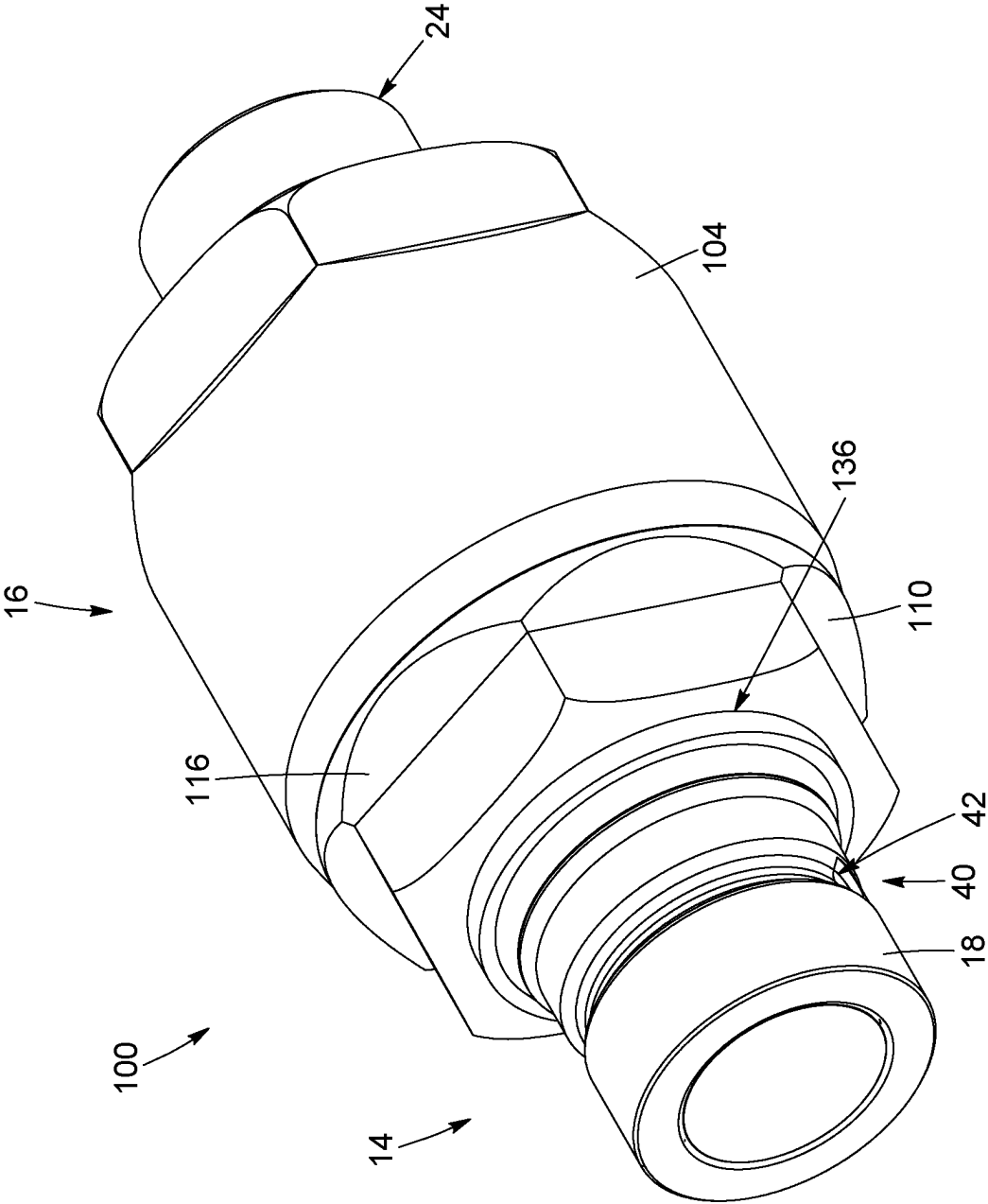
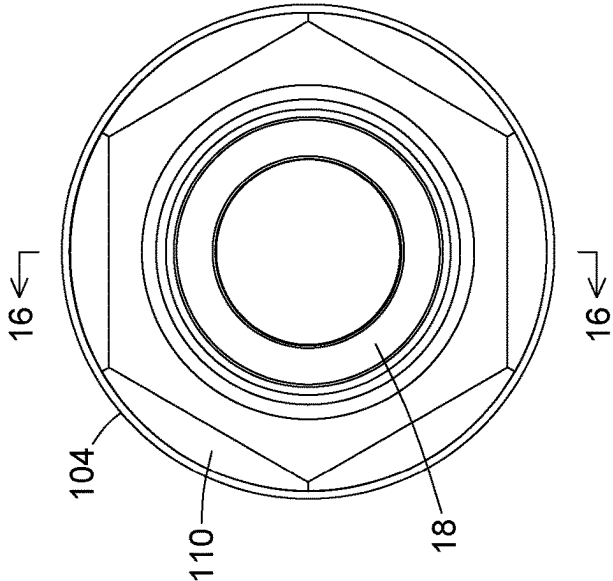
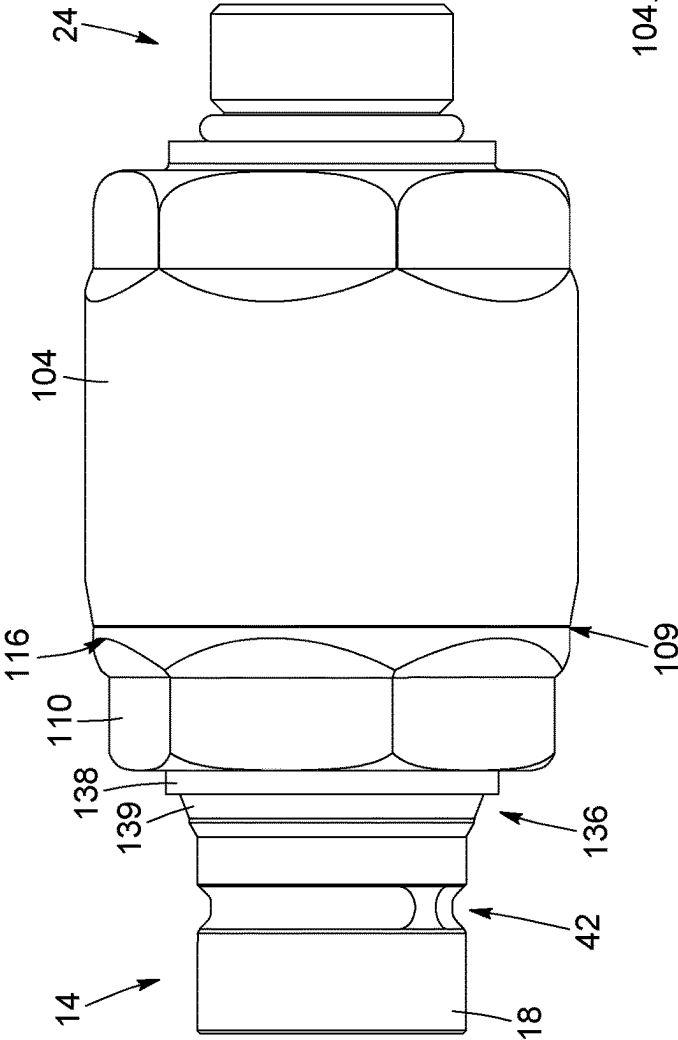


FIG. 13



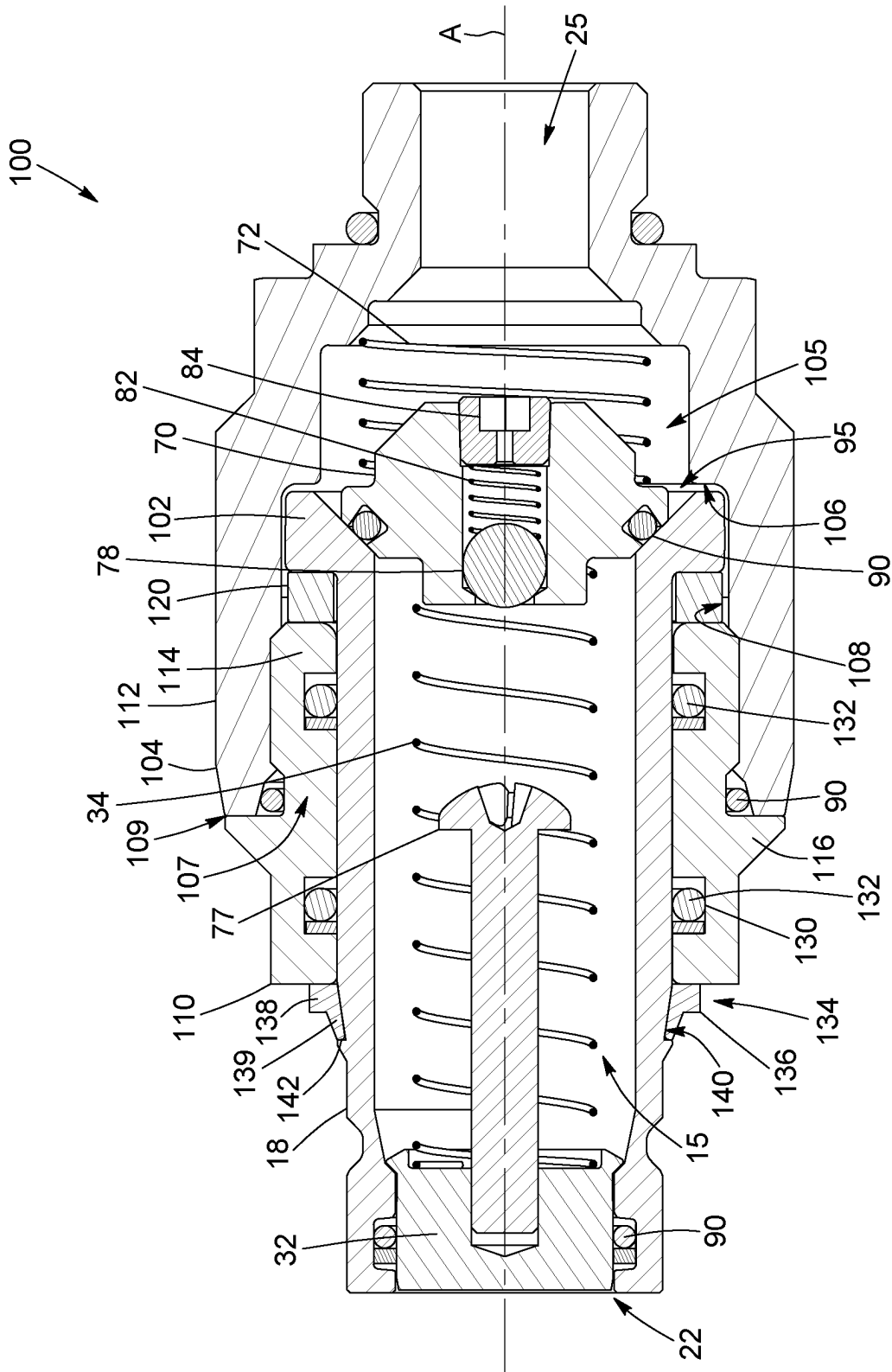


FIG. 16

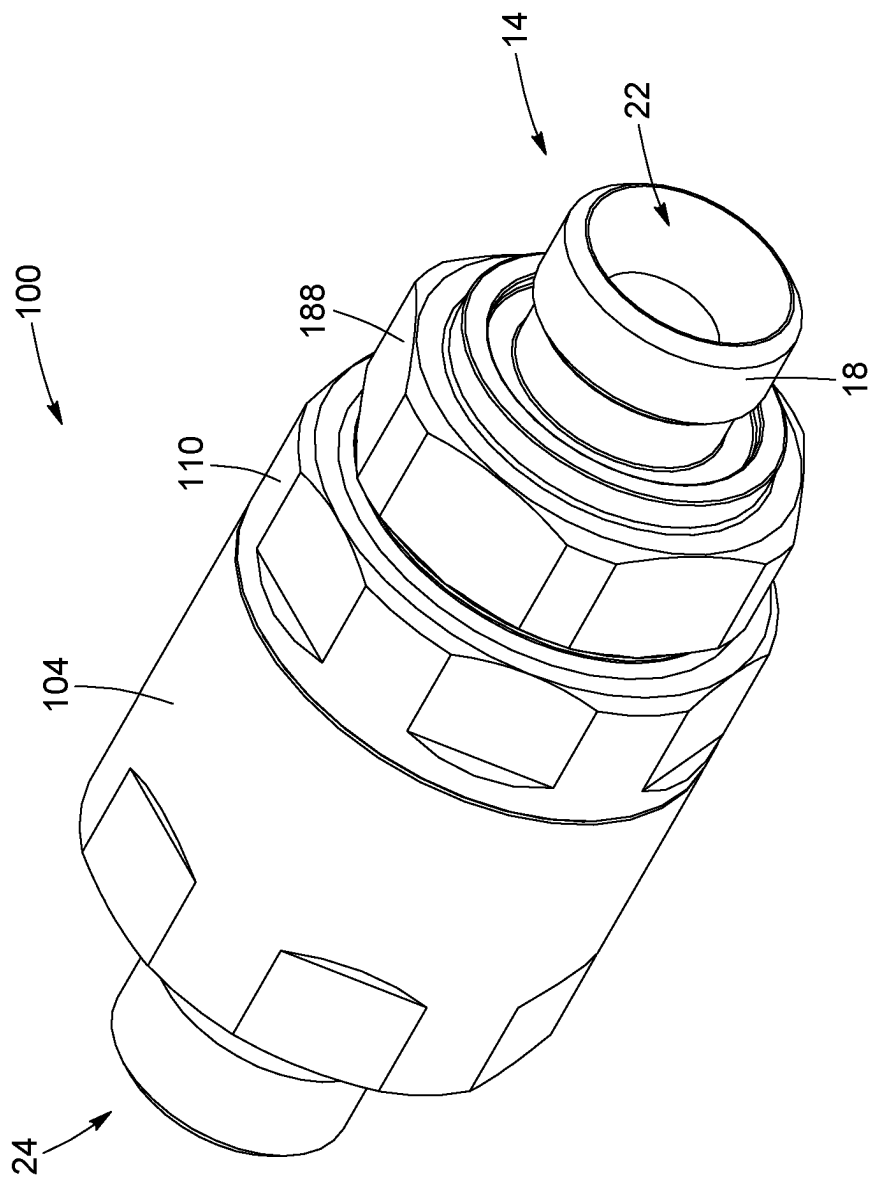


FIG. 17

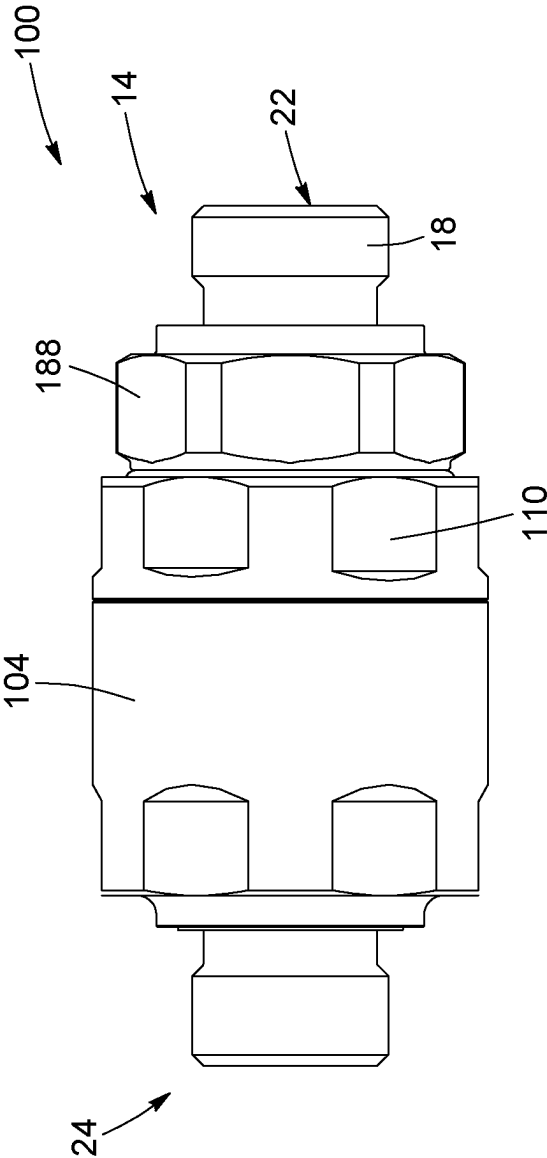


FIG. 18

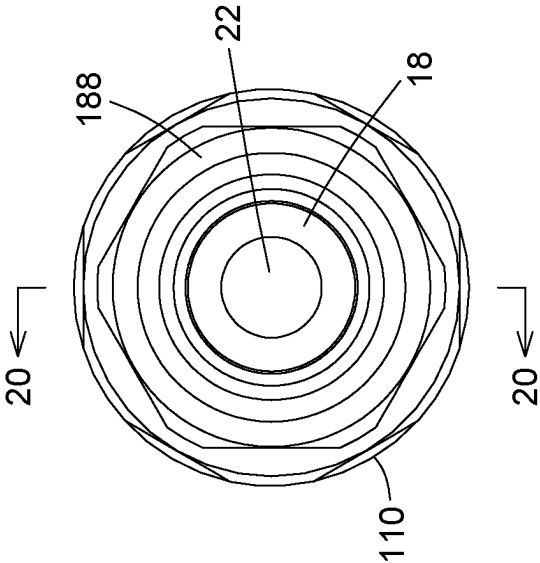


FIG. 19

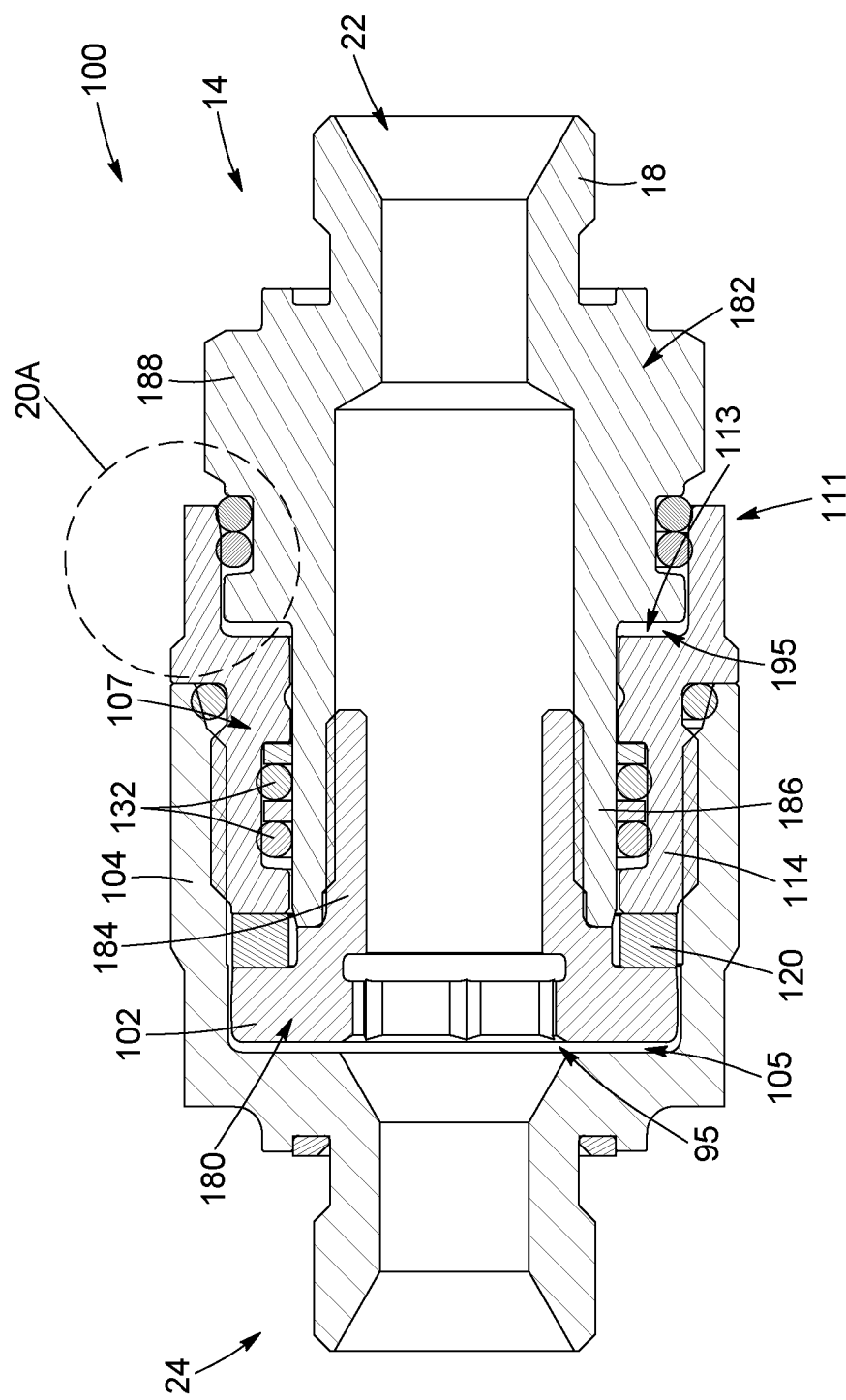


FIG. 20

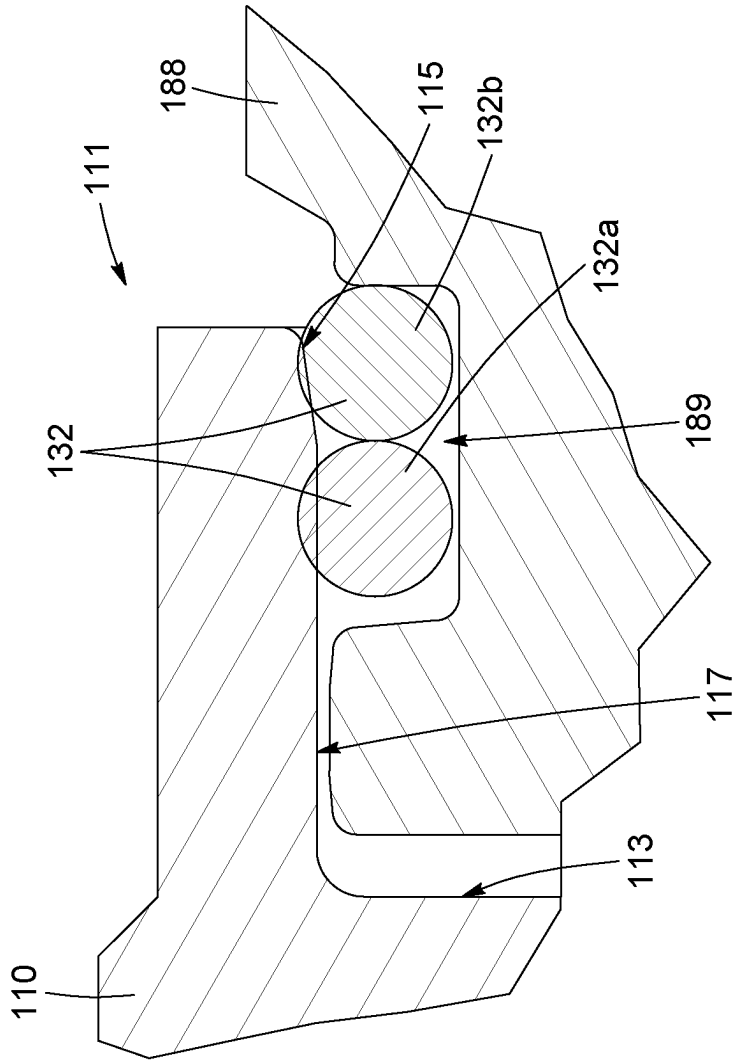


FIG. 20A

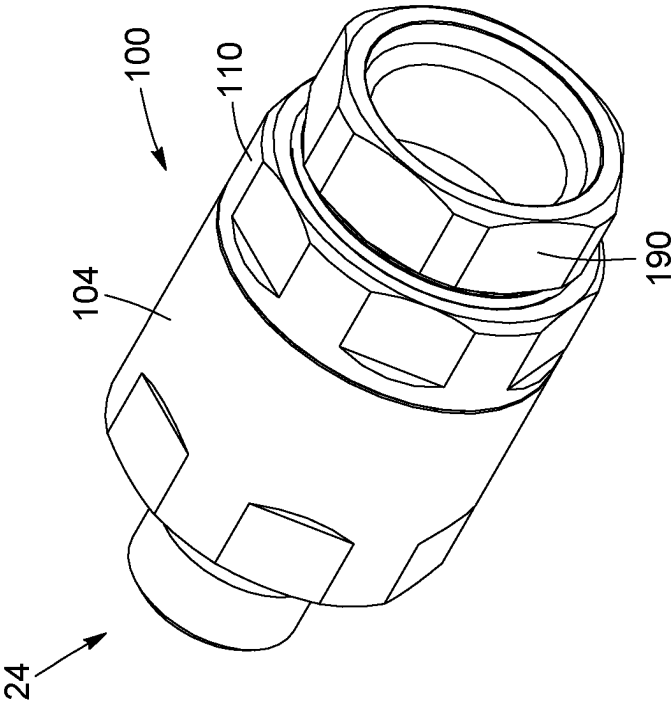


FIG. 21

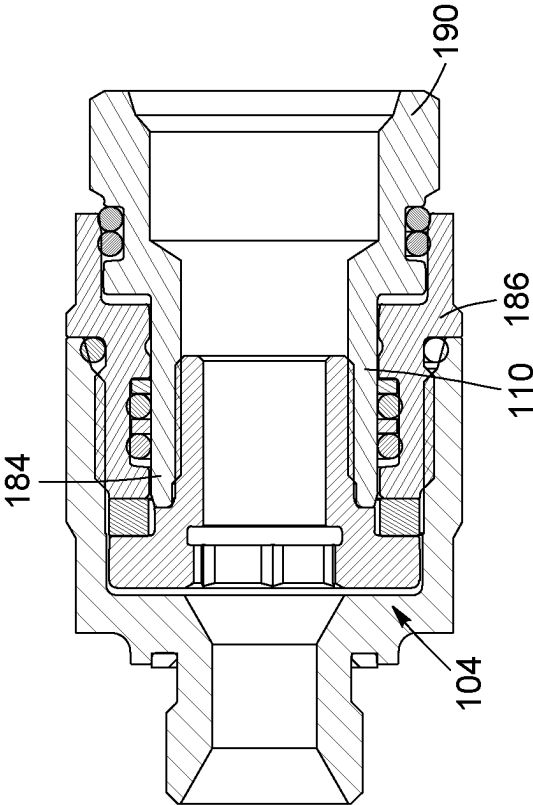


FIG. 22

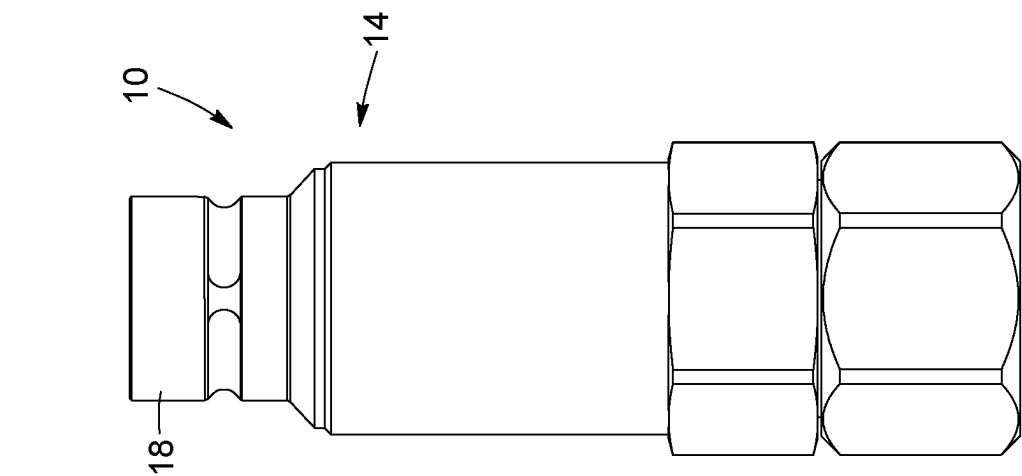


FIG. 23

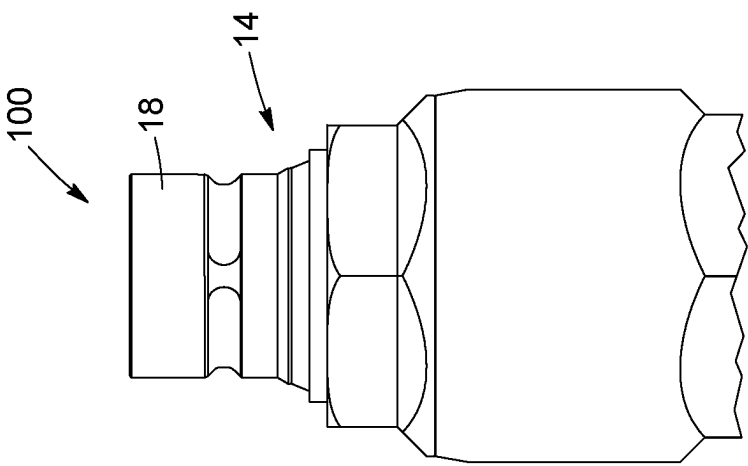


FIG. 24

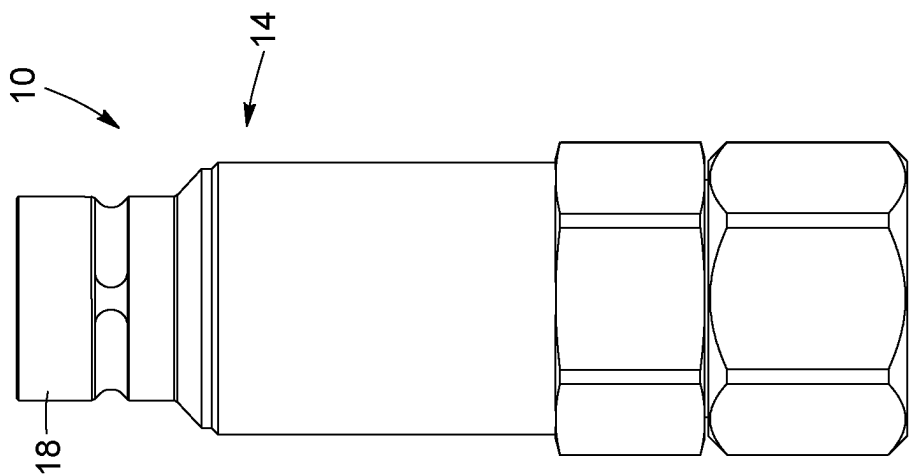


FIG. 25

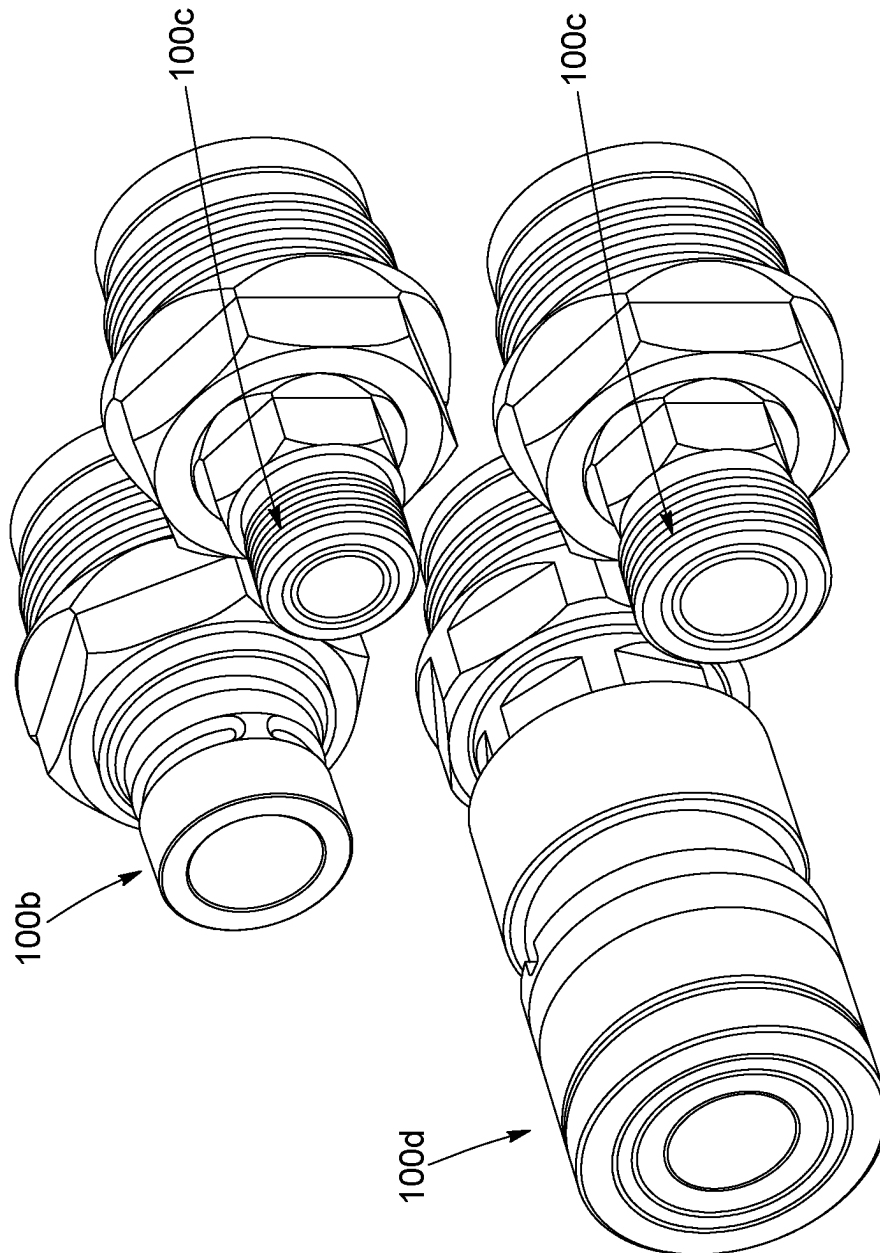


FIG. 25B

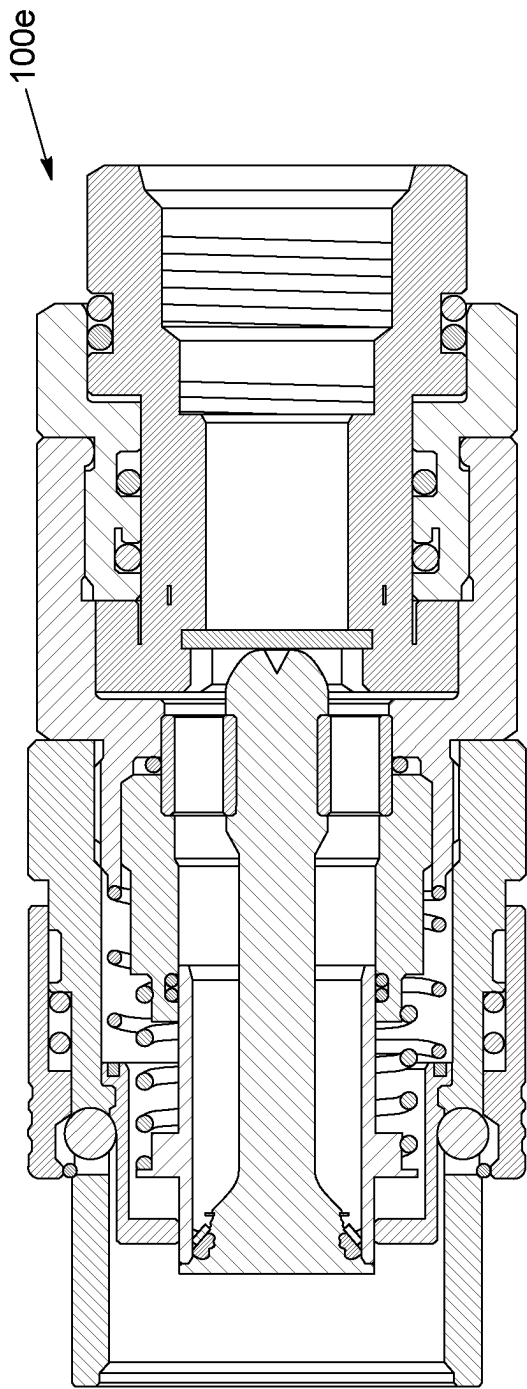


FIG. 25C

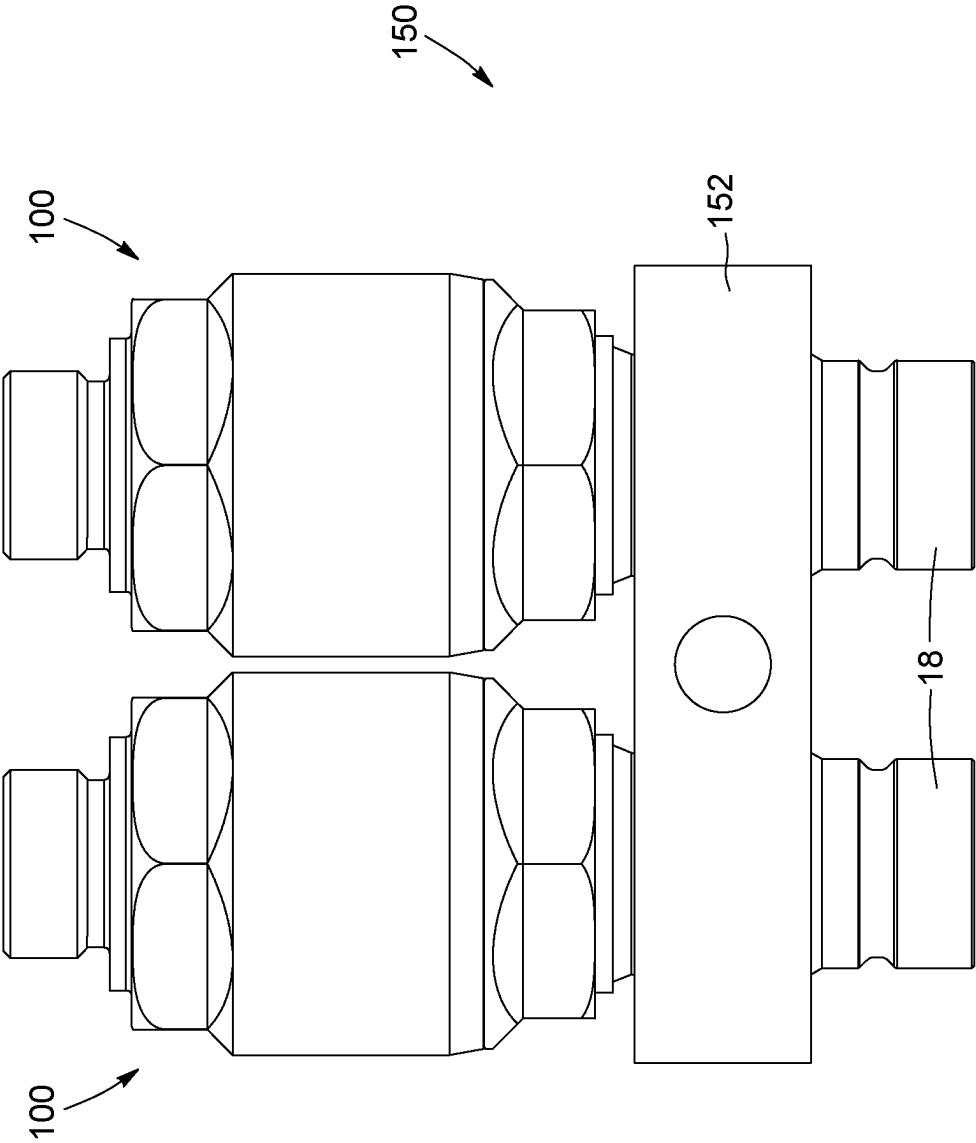
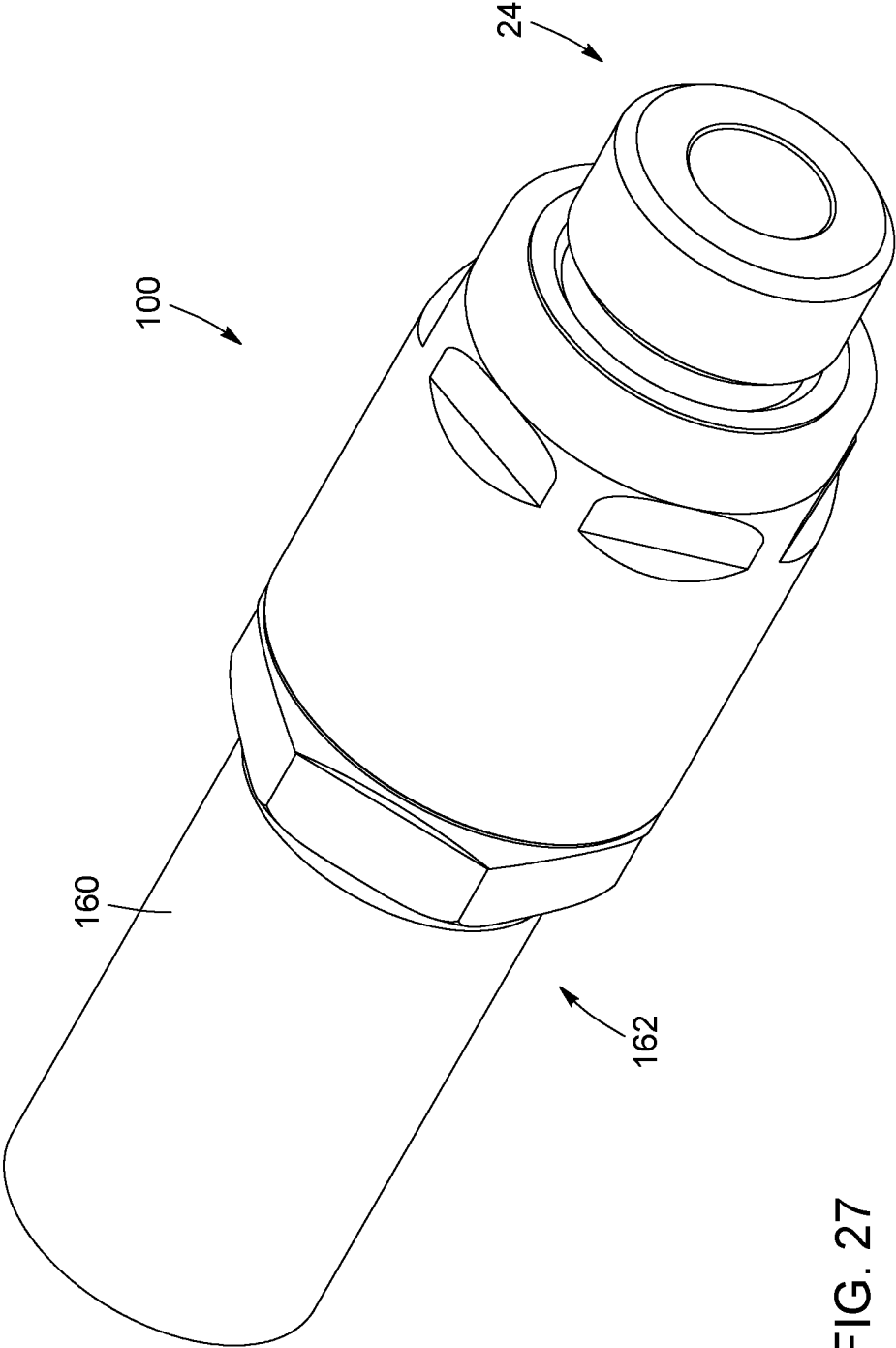
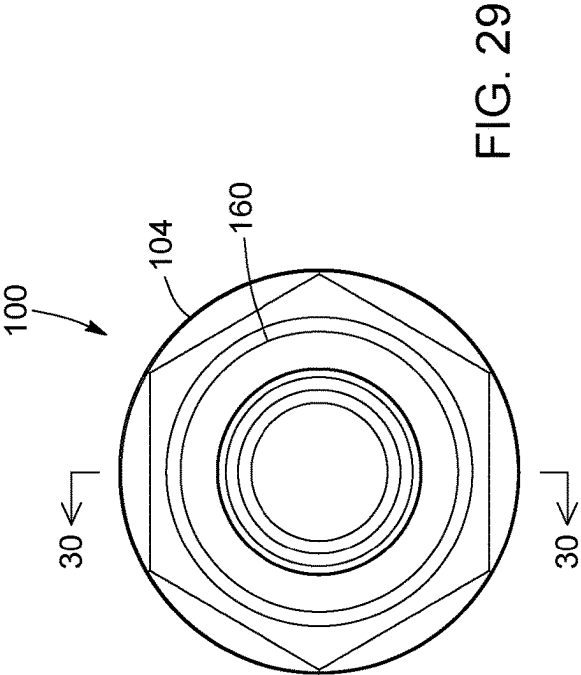
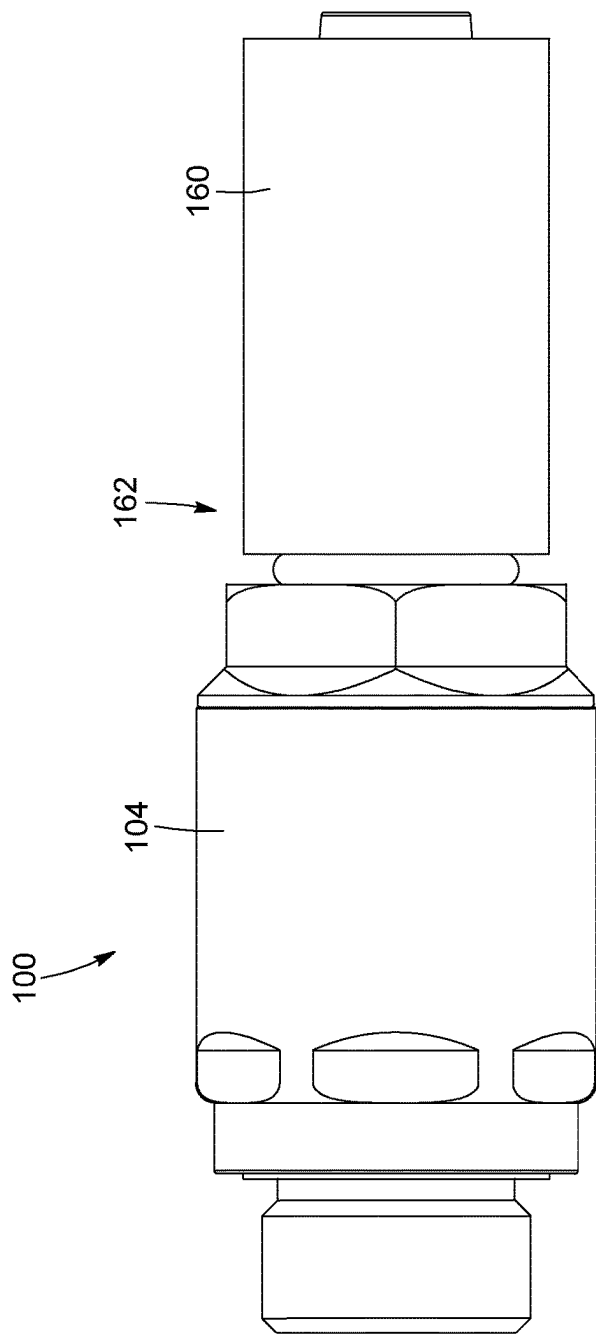


FIG. 26





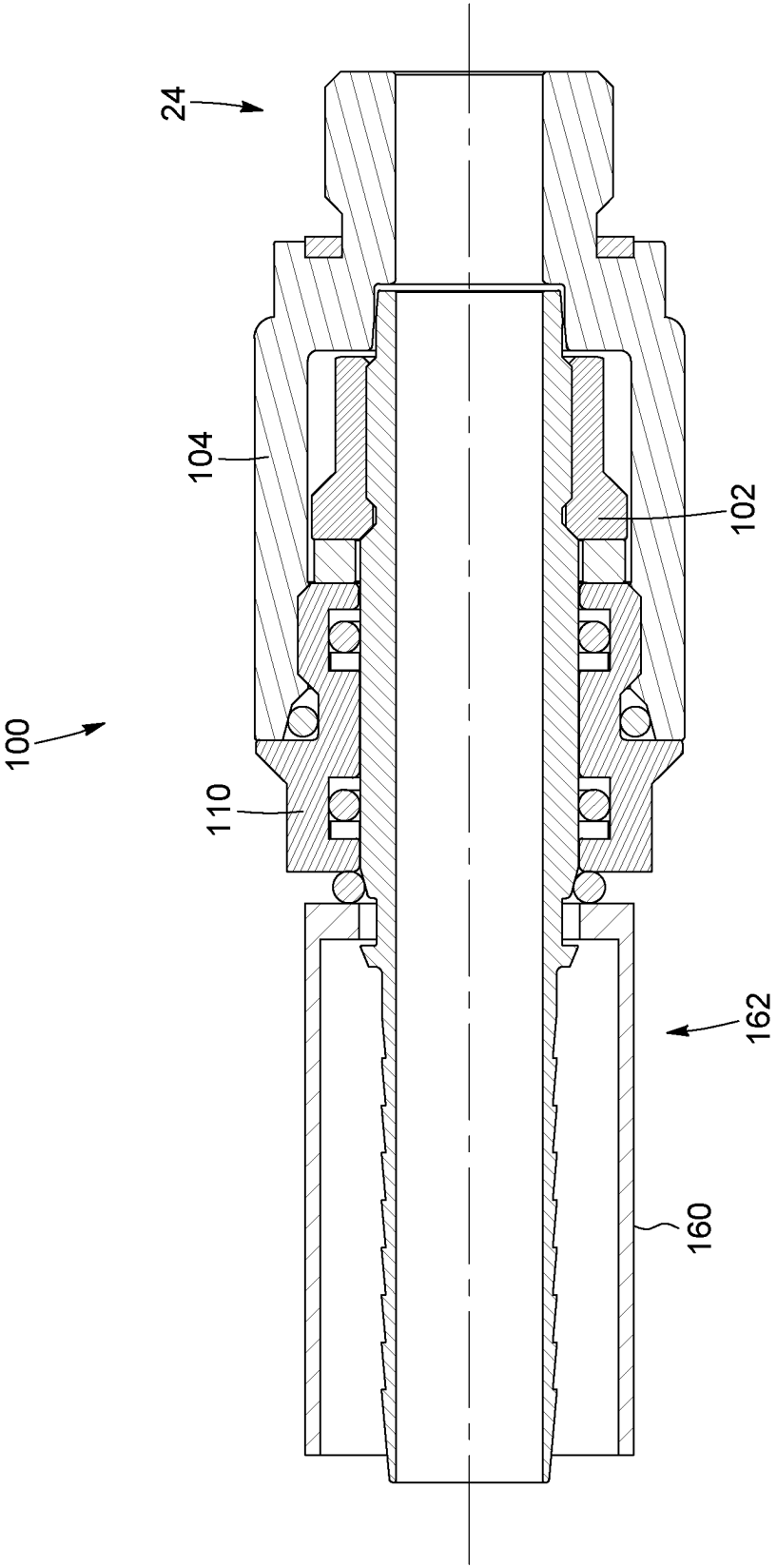


FIG. 30

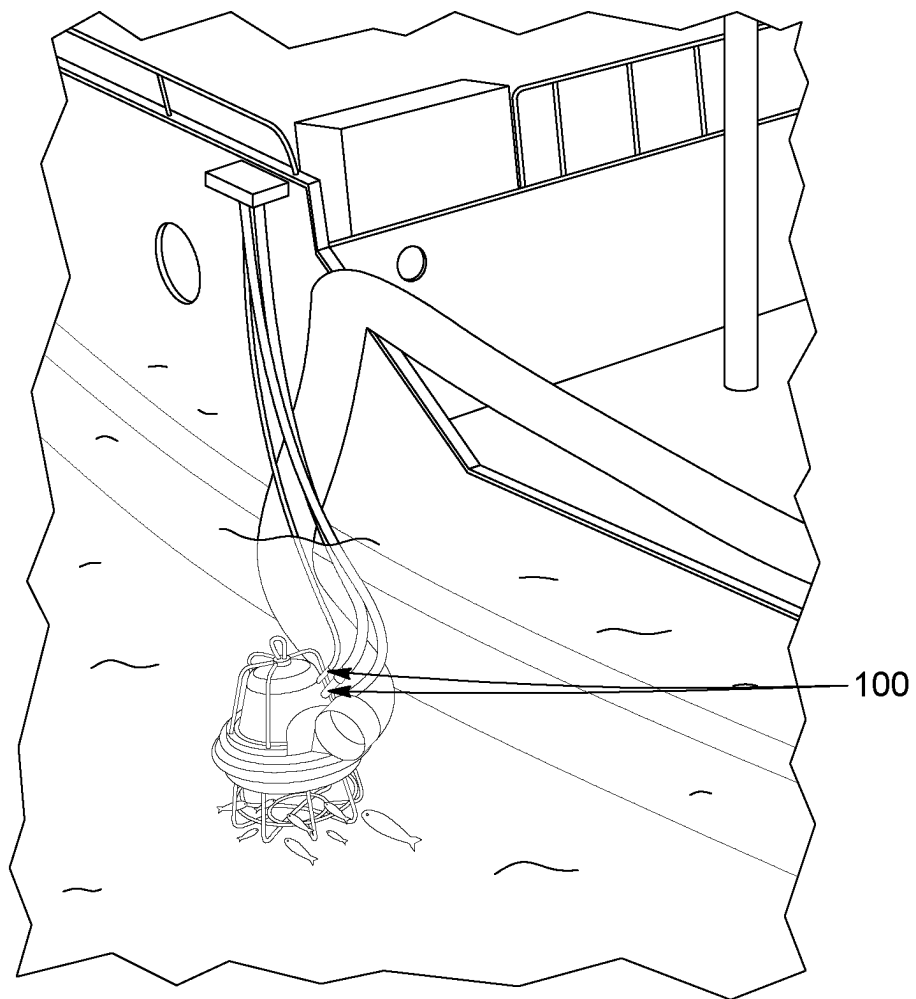


FIG. 31

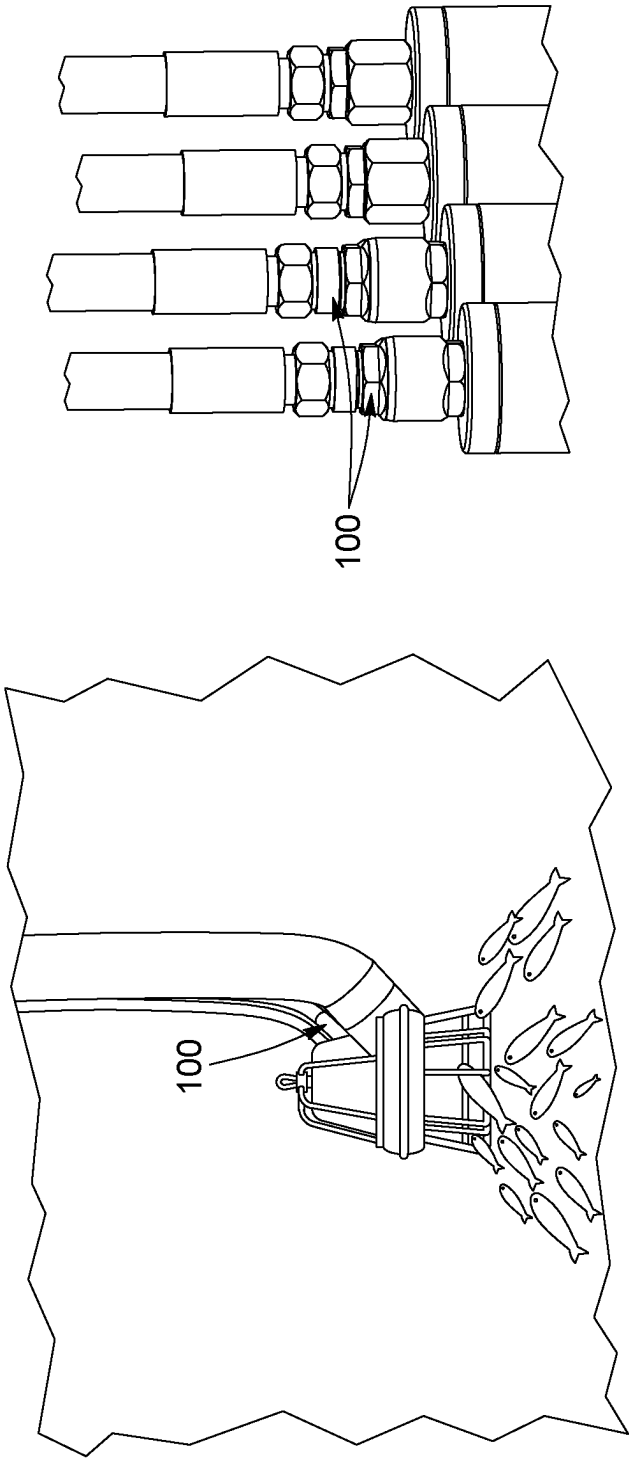


FIG. 32

FIG. 33

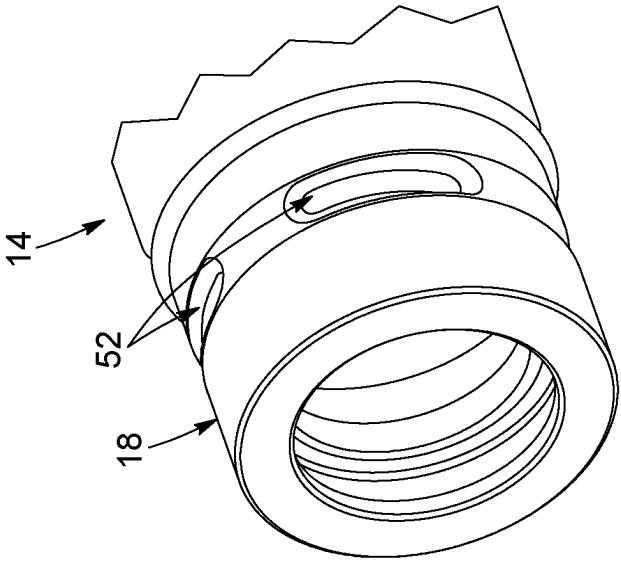


FIG. 34

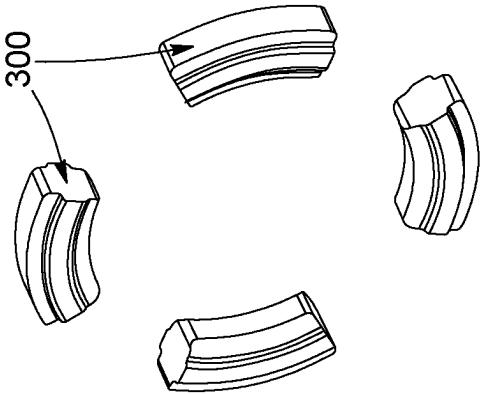


FIG. 35

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QUICK LOCK RELEASE COUPLING**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a National Stage of International patent application PCT/CA2022/050686, filed on May 3, 2022, which claims priority to U.S. Provisional Patent Application Nos. 63/201,573 and 63/201,574 filed May 5, 2021, the disclosures of which are incorporated by reference in their entireties.

TECHNICAL FIELD

The present disclosure generally relates to couplings, and more particularly, to quick lock release couplings for use with hydraulic lines.

BACKGROUND

Several types of couplings are used in the fluid transfer industry. For example, push-to-connect connections, or “quick-release couplings”, are different even though the applications and some of the structure are similar to other known couplings. Widely used in the pneumatic industry and for low-pressure applications, quick-release couplings are not designed for higher-pressure applications, such as for the transfer of fluids on heavy machinery where pressurized fluid are flowing and where torsion constraints are present. Hydraulic lines, for instance, being for the most part constructed with braided or intertwined metallic wires or synthetic plastic material, are quite sensitive to torsion constraints.

Swivelling couplings are components that enable two or more hydraulic lines to be interconnected axially or angularly, and to reduce torsion constraints that could affect their physical integrity or internal structure. Quick-release couplings often consist of a male connector with a groove around its periphery allowing the ball bearings of the female connector to be trapped in said groove. Thus, the axial movement is eliminated, while the assembly is free to rotate. Therefore, it becomes similar in every way to a rotary joint consisting of one or more rows of balls and a seal. Pressurized hydraulic fluid, along with the rotation of known rotary joints and swivelling couplings, accelerates wear and tear of the various seals within the coupling, which requires frequent maintenance, repair and replacement.

SUMMARY

According to an aspect, a quick release coupling for providing a connection between a first fluid line provided with a female component comprising a ball bearing assembly and a second fluid line is provided. The quick release coupling includes a male component having a fluid passage therethrough and comprising a body portion having an internal surface at least partially defining the fluid passage; and a shank portion extending from the body portion and being adapted to engage the female component of the first fluid line, the shank portion having a shank portion port adapted to establish fluid communication between the first fluid line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly and block a rotational movement and an axial movement of the male component relative to the first fluid line when engaging the first fluid line. The quick release coupling also includes a coupling adapter having a fluid

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channel therethrough and being securable to the male component proximate the body portion to establish fluid communication between the fluid channel and the fluid passage, the coupling adapter being adapted to engage the second fluid line and comprising an adapter port adapted to establish fluid communication between the second fluid line and the fluid channel.

According to a possible embodiment, the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.

According to a possible embodiment, the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking the axial movement of the male component.

According to a possible embodiment, the groove extends circumferentially about the shank portion in a single plane.

According to a possible embodiment, the groove extends partially about the shank portion such that the groove comprises a first end and a second end spaced from one another and defining an ungrooved portion therebetween.

According to a possible embodiment, the ungrooved portion is shaped and adapted to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.

According to a possible embodiment, the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.

According to a possible embodiment, the detent comprises a plurality of recesses provided along the groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.

According to a possible embodiment, the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.

According to a possible embodiment, the protrusion extends from within the groove.

According to a possible embodiment, the protrusion is integrally formed with the shank portion or is welded thereto.

According to a possible embodiment, the quick release coupling further includes a valve assembly operable to control fluid flow along the fluid passage of the male component before, during and after engaging the shank portion with the first fluid line.

According to a possible embodiment, the valve assembly comprises a valve head provided within the fluid passage proximate the shank portion port and being operable in a closed position where fluid flow is blocked through the shank portion port, and an open position, the valve assembly further comprising a head spring connected to the valve head and being adapted to bias the valve head in the closed position.

According to a possible embodiment, the valve assembly comprises a valve body provided within the fluid passage along the body portion and being operable in a closed position where fluid flow through the fluid passage is blocked, and an open position where fluid flow through the fluid passage is allowed, the valve assembly further comprising a body spring connected to the valve body and being adapted to bias the valve body in the closed position.

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According to a possible embodiment, the coupling adapter further comprises a housing having internal surfaces comprising at least one radial surface and at least one axial surface, said internal surfaces defining a cavity having an open end, wherein the body portion of the male component is shaped and sized to engage the cavity via the open end and abut the at least one radial surface.

According to a possible embodiment, the quick release coupling further includes a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the body portion within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivelably interconnected.

According to a possible embodiment, the quick release coupling further includes a thrust washer surrounding the shank portion between the nut and the body portion.

According to a possible embodiment, the shank portion comprises a flange extending radially outward and having a flange surface being substantially perpendicular to the external surface of the shank portion and facing the nut, and wherein the quick release coupling further comprises an outer sealing element surrounding the shank portion and extending between the flange surface and the nut.

According to another aspect, a male component of a quick release coupling for connection with a hydraulic line provided with a female component comprising a ball bearing assembly is provided, with the male component having a fluid passage therethrough and comprising a shank portion extending along a longitudinal axis and being adapted to engage the hydraulic line, the shank portion having a shank portion port adapted to establish fluid communication between the hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly to block a rotation of the male component about the longitudinal axis.

According to a possible embodiment, the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.

According to a possible embodiment, the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking an axial movement of the male component along the longitudinal axis.

According to a possible embodiment, the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.

According to a possible embodiment, the detent comprises a plurality of recesses provided along the groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.

According to a possible embodiment, the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls to block the rotational movement of the male component.

According to a possible embodiment, the protrusion extends from within the groove.

According to a possible embodiment, the protrusion is integrally formed with the shank portion or is welded thereto.

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According to another aspect, a quick release coupling for providing a connection between a pair of hydraulic lines is provided. The quick release coupling includes a male component having a fluid passage therethrough and comprising a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line. The quick release coupling also has a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being adapted to engage a second hydraulic line and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.

According to another aspect, a use of a quick release coupling for providing a connection between a pair of hydraulic lines is provided. The quick release coupling includes a male component having a fluid passage therethrough and comprising a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line. The quick release coupling also includes a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being adapted to engage a second hydraulic line and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.

According to a possible embodiment, the quick release coupling is adapted to operate between about 0 psi and 5000 psi.

According to another aspect, a use of the quick release coupling defined above for providing a connection between a pair of hydraulic lines is provided.

According to another aspect, a method of connecting a first fluid line to a second fluid line using a quick release coupling as defined above is provided. The method includes connecting the coupling adapter to the second fluid line; and engaging the shank portion with the first fluid line to block a rotational movement and an axial movement of the quick release coupling relative to the first fluid line.

According to another aspect, a method of connecting a first fluid line provided with a female component to a second fluid line is provided. The method includes connecting a first end of a hydraulic coupling to the second fluid line; and connecting a second end of the hydraulic coupling to the first fluid line, the second end being provided with a male component configured to engage the female component and block a rotational movement and an axial movement of the male component relative to the female component.

According to a possible embodiment, the hydraulic coupling is a quick release coupling.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a quick lock release coupling provided with a restraining mechanism for pre-

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venting movement of a portion of the quick lock release coupling, according to a possible embodiment.

FIG. 2 is a side view of the quick lock release coupling shown in FIG. 1, showing a male component and a coupling adapter extending at opposite ends of the quick lock release coupling, according to an embodiment.

FIG. 3 is a front view of the quick lock release coupling shown in FIG. 1, showing a valve head provided within a port of the coupling, according to an embodiment.

FIG. 4 is a sectional view of the quick lock release coupling shown in FIG. 3, taken along line 4-4, showing a fluid passage extending through a housing of the quick lock release coupling, according to an embodiment.

FIGS. 5 to 8 are top views of various embodiments of a male component, showing possible embodiments of the restraining mechanism.

FIG. 9 is a perspective view of a quick lock release coupling, according to an alternate embodiment.

FIG. 10 is a side view of the quick lock release coupling shown in FIG. 9, showing a male component and a coupling adapter extending at opposite ends of the quick lock release coupling, according to an embodiment.

FIG. 11 is a front view of the quick lock release coupling shown in FIG. 9, showing a valve head provided within a port of the coupling, according to an embodiment.

FIG. 12 is a sectional view of the quick lock release coupling shown in FIG. 11, taken along line 12-12, showing a pressure eliminator provided within a valve body, according to an embodiment.

FIG. 13 is a perspective view of a swivelling coupling, according to an alternate embodiment.

FIG. 14 is a side view of the swivelling coupling shown in FIG. 13, showing the male component coupled within the coupling adapter via a nut, according to an embodiment.

FIG. 15 is a front view of the swivelling coupling shown in FIG. 13, showing a valve head provided within the port of the male component, according to an embodiment.

FIG. 16 is a sectional view of the swivelling coupling shown in FIG. 15, taken along line 16-16, showing a flange portion of the male component provided between internal surfaces of the coupling adapter housing and the nut, according to an embodiment.

FIG. 17 is a perspective view of a swivelling coupling, according to an alternate embodiment.

FIG. 18 is a side view of the swivelling coupling shown in FIG. 17, showing the male component coupled within the coupling adapter via a nut, according to an embodiment.

FIG. 19 is a front view of the swivelling coupling shown in FIG. 17, showing a male component port, according to an embodiment.

FIG. 20 is a sectional view of the swivelling coupling shown in FIG. 19, taken along line 20-20, showing the shank portion of the male component being coupled to the flange portion provided within the coupling adapter housing, according to an embodiment.

FIG. 20A is an enlarged view of a portion of the swivelling coupling shown in FIG. 20, showing a pair of sealing rings provided about the shank portion between the shank portion and the nut, according to an embodiment.

FIG. 21 is a perspective view of a swivelling coupling, according to an alternate embodiment.

FIG. 22 is a sectional view of the swivelling coupling shown in FIG. 21, showing a shank portion defining a female connection, according to an embodiment.

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FIGS. 23 to 25 are top views of couplings, showing a coupling cartridge (FIG. 23), a swivelling coupling (FIG. 24) and a quick release male coupling (FIG. 25) according to possible embodiments.

FIG. 25B is a perspective view of a multiport swivel assembly provided with various implementations of couplings, according to an embodiment.

FIG. 25C is a sectional view of an alternative embodiment of a coupling, showing a threaded connection port.

FIG. 26 is a top view of a pair of swivelling couplings coupled to a multiport swivel assembly, according to an embodiment.

FIG. 27 is a perspective view of a coupling provided with a crimp fitting, according to an alternate embodiment.

FIG. 28 is a side view of the coupling shown in FIG. 27, showing the coupling adapted and the crimp fitting extending at opposite ends of the coupling, according to an embodiment.

FIG. 29 is a front view of the coupling shown in FIG. 27, according to an embodiment.

FIG. 30 is a sectional view of the quick lock release coupling shown in FIG. 29, taken along line 30-30, showing a fluid passage extending through a housing of the quick lock release coupling, according to an embodiment.

FIGS. 31 and 32 are representations of a possible application comprising one or more couplings, showing a fishing vessel pump prior to being submerged (FIG. 31) and the fishing vessel pump underwater for operation (FIG. 32).

FIG. 33 is an illustration of a hydraulic machine comprising couplings installed in a vertical configuration, according to a possible embodiment.

FIGS. 34 and 35 are perspective views of an alternate embodiment of a coupling, showing latching elements (FIG. 35) having an elongate shape and extending partially circumferentially within ports of a male component (FIG. 34).

DETAILED DESCRIPTION

As will be explained below in relation to various implementations, the present disclosure describes devices and systems for providing improved connections for hydraulic machinery, such as quick lock release couplings configured to be coupled to and between hydraulic lines. The present disclosure describes a quick lock release coupling that is easily and reliably connectable to hydraulic lines adapted for the transport of hydraulic fluid at elevated pressures. The quick lock release coupling can include a male component shaped and sized to engage a female component of a hydraulic line. The male component includes a restraining mechanism adapted to block movement of the male component relative to the hydraulic line (e.g., relative to the female component) when coupled thereto. The female component traditionally includes a ball bearing assembly adapted to prevent axial movement of the male component while enabling rotation of the male component when coupled thereto. The restraining mechanism described herein can be further adapted to block a rotational movement of the male component. For example, the restraining mechanism can include a detent shaped and sized to extend between a pair of adjacent balls of the ball bearing assembly, thereby preventing rotation of the male component.

It is thus noted that the male component is adapted to remain generally static relative to the hydraulic line due to the restraining mechanism. Moreover, it is appreciated that blocking movement of the male component relative to the hydraulic line can improve the lifespan and efficiency of the various sealing components of the quick lock release cou-

pling, which sustain less stress, less friction, less compression forces, etc., due to the static assembly of the male component.

The quick lock release couplings can be configured to enable the transport of hydraulic fluid for various operations. The couplings can be implemented on various apparatuses, machines and devices, such as forestry combo harvesters, industrial reach mowers, compact tracked loader attachments, agriculture attachments, among others. As will be described herein, the hydraulic coupling can be adapted to provide interconnection between two fluid lines. The coupling can enable a coaxial interconnection of two fluid lines, but it should be understood that the coupling may be adapted to provide a different interconnection, such as a 90° connection or an oblique connection. Alternatively, or additionally, the quick lock release coupling can include an integrated swivel assembly adapted to enable motion of the one or more hydraulic lines connected to the quick lock release coupling and provide long-lasting sealing capabilities before servicing.

With reference to FIGS. 1 to 4, an example quick release coupling 10 (or simply “coupling”) for interconnecting and establishing fluid communication between a pair of fluid lines is illustrated. The coupling 10 includes one or more components connected to one another and defining a central passage 12 through the coupling 10. Therefore, fluid, such as hydraulic fluid, can flow through the coupling 10 via the central passage 12. In some embodiments, the coupling 10 includes a first component configured to engage a first fluid line, and a second component configured to engage a second fluid line, thereby connecting the fluid lines to one another. The first and second components are secured to one another and each define a portion of the central passage 12. In this embodiment, the central passage 12 is generally longitudinal, with the inlet of the first component being substantially aligned with the inlet of the second component. However, it is appreciated that other configurations are possible, such as having a curved or bent central passage 12.

In this embodiment, the first component of the coupling 10 can include a male component 14 defining a fluid passage 15 therethrough, and having a body portion 16 and a shank portion 18 extending from a first side of the body portion 16. As seen in FIG. 4, the male component 14 has internal surfaces 20 extending along the body and shank portions 16, 18 which define the fluid passage 15. Furthermore, and as will be described further below, the shank portion 18 is adapted to engage the first fluid line and includes a first port, or male component port 22, configured to establish fluid communication between the first fluid line and the fluid passage 15.

In this embodiment, the second component of the coupling 10 includes a coupling adapter 24 defining a fluid channel 25 therethrough. The coupling adapter 24 is securable to the male component 14 in a manner such that the fluid channel 25 communicates with the fluid passage 15. It is noted that, in this embodiment, the combination of the fluid passage 15 and the fluid channel 25 forms the central passage 12 of the coupling 10, although other configurations are possible. In this embodiment, the coupling adapter 24 is adapted to be secured to the body portion 16 of the male component 14, opposite the shank portion 18. For example, the body portion 18 and the coupling adapter 24 can have complementarily shaped threads, enabling the coupling adapter 24 to be threaded into the body portion 18. In some embodiments, the coupling 10 includes an adapter seal 28 provided between the coupling adapter 24 and male component 14 to further secure the components together and

prevent fluid from leaking therebetween. As seen in FIGS. 2 and 4, the coupling adapter 24 is provided with an adapter port 26 adapted to engage the second fluid line and establish fluid communication between the second fluid line and the fluid channel 25. It is thus noted that the first fluid line and the second fluid line are fluidly connected via the coupling 10, where fluid flows through the fluid channel 25 and fluid passage 15, in either direction.

Still referring to FIGS. 1 to 4, the coupling 10 can be provided with a valve assembly 30 configured to control the flow of fluid through the coupling 10 (e.g., through the male component 14, through the coupling adapter 24 or a combination thereof). The valve assembly 30 can be operable between a closed configuration, where fluid flow through the first port 22 is blocked, and an open configuration, wherein fluid flow through the first port 22 is allowed. In this embodiment, the valve assembly 30 includes a valve head 32 provided along the fluid passage 15, and more specifically within the first port 22. The valve head 32 is adapted to move between a closed position (seen in FIG. 4) where the valve head 32 obstructs the first port 22, and an open position, where the valve head 32 is displaced such that the first port 22 becomes unobstructed.

In this embodiment, the valve assembly 30 includes a resilient element 34 coupled to the valve head 32 and configured to bias the valve head 32 in the closed position. For example, the resilient element 34 can include a spring 35 operatively connected to the valve head 32 to bias the valve head 32 in the closed position. As seen in FIG. 4, the spring 35 can extend along the fluid passage 15 and abut against an internal surface of the coupling adapter 24. It is noted that opening the valve assembly 30 (i.e., moving the valve head 32 in the open position) includes moving the valve head 32 further within the fluid passage 15 (i.e., towards the coupling adapter 24), thereby compressing the spring 35 and opening the first port 22. In some embodiments, the female component can be provided with an actuator, such as a stem, configured to push against the valve head 32 when engaging the shank portion 18 with the first fluid line. In other words, while being pressed against the stem of the female component, the valve head retreats inside the male component, thereby opening the first port 22. The valve head 32 illustratively has a flat outer surface 36 adapted to be generally coplanar with the end of the shank portion 18 when in the closed position.

In some embodiments, the shank portion 18 includes a restraining mechanism 40 adapted to prevent undesired or accidental disengagement of the male component 14 from the first fluid line. For example, the restraining mechanism 40 can be provided along the external surface of the shank portion 18 for engaging an inner surface of the first fluid line. Traditionally, the first fluid line can include a female component (not shown) provided with a ball bearing assembly installed circumferentially along the inner surface. Moreover, and as seen in FIGS. 1, 2 and 4, the restraining mechanism 40 includes a groove 42 extending circumferentially about the shank portion 18, and being shaped and sized to receive the ball bearing assembly therein. As such, once the shank portion 18 engages the first fluid line, the ball bearing assembly and the groove 42 cooperate with one another to block axial movement of the male component 14 relative to the first fluid line, thereby preventing disengagement therefrom.

In some embodiments, the restraining mechanism 40 is further adapted to block a rotational movement of the male component 14 relative to the first fluid line. More particularly, it is noted that the ball bearing assembly can enable

rotation of the male component **14** about its longitudinal axis (A). However, the restraining mechanism **40** can include a detent **44** shaped and sized to engage the ball bearing assembly to prevent rotation of the male component **14**. The detent **44** can be fixedly connected to the shank portion **18** such that engagement of the detent **44** with the ball bearing assembly prevents rotation of the shank portion **18** (and thus of the male component **14**) in either direction about the longitudinal axis. As seen in FIG. 2, the detent **44** can include a protrusion **46** extending from the shank portion **18** and which is shaped and sized to extend between a pair of adjacent ball bearings to block the rotation of the male component relative to the first fluid line.

In this embodiment, the protrusion **46** extends from within the groove **42** to facilitate its engagement with the ball bearing assembly (e.g., between a pair of adjacent ball bearings). Furthermore, the protrusion **46** can be integrally formed as part of the shank portion **18** (e.g., during machining of the groove **42**), although it is appreciated that other configurations are possible. For example, and with reference to FIGS. 5 to 8, the detent **44** can be provided about the shank portion **18** between the body portion **16** and the groove **42** (see FIGS. 5 and 6), or within the groove **42** (see FIGS. 7 and 8). In the embodiment of FIG. 5, the detent **44** can include a ball **48**, or dome-shaped element, connected to and extending from the shank portion **18** to engage the female component of the first fluid line. It should be noted that the female component may require an additional complementarily shaped element configured to engage with the ball **48** to block the rotational movement of the male component. The ball **48** can be spot welded onto the shank portion **18**, integrally formed therewith or connected thereto using any suitable method.

Alternatively, and with reference to FIG. 6, the detent **44** can include a protuberance **50** provided between the body portion **16** and the groove **42** along the external surface of the shank portion **18**. In a similar fashion as the ball **48**, the protuberance **50** is configured to engage a complementarily-shaped element of the female component for blocking the rotation of the male component. In some embodiments, the complementarily-shaped element includes a recess (not shown) having a shape conforming to the shape of the protuberance **50** such that the male component **14** is adapted to engage the female component in a predetermined orientation. It is appreciated that the male component **14** can include a plurality of detents **44** provided about the shank portion **18**, both in and/or proximate the groove **42**.

In yet another embodiment, and as seen in FIG. 7, the detent **44** can include one or more recesses **52** provided about the shank portion **18** and/or within the groove **42**. The recesses **52** are shaped and sized to receive a ball bearing of the ball bearing assembly such that rotation of the male component **14** about the longitudinal axis is blocked. It is noted that the ball bearing assembly includes a finite number of ball bearings, and that the detent **44** can include any suitable number of recesses **52**, such as fewer than the number of ball bearings, more than the number of ball bearings, or the same number. For example, the ball bearing assembly can include twelve (12) ball bearings, and the detent **44** can include twenty-four (24) recesses **52** to facilitate engagement of the ball bearings with the detent.

Now referring more specifically to FIG. 8, in this embodiment, the detent **44** includes the protrusion **46** provided along the groove **42**, and further includes the ball **48** connected to and extending from the protrusion **46**. The protrusion **46** can be adapted to prevent rotation of the male component **14** when engaged with the female component,

while the ball **48** can facilitate positioning the male component **14** to have a pair of adjacent ball bearings on either side of the protrusion **46**. It should be noted that the detent **44** of the restraining mechanism **40** can include any suitable component, feature or combination thereof to prevent rotation of the male component **14**. For example, the detent **44** can include the recesses **52** provided along the groove **42**, in addition to having a protuberance **50** extending from the body portion **18** and between a pair of adjacent ball bearings. It is appreciated that any other combination of the detent **44** can be used and is possible.

Now referring to FIGS. 9 to 12, another embodiment of the coupling **10** will be described. The shank portion **18**, and related components (e.g., the restraining mechanism **40**) can be substantially the same as the previously described embodiment, and adapted to engage the first fluid line, and prevent axial and rotational movement of the male component. In this embodiment, the body portion **16** is elongated and includes internal surfaces adapted to define one or more internal cavities **60** along the fluid passage **15**. Referring more specifically to FIG. 12, the internal cavities **60** can include a first cavity **62** proximate the shank portion **18**, and a second cavity **64** proximate the coupling adapter **24** when the coupling adapter **24** is connected to the male component **14**.

In this embodiment, the valve assembly **30** further includes a valve body **70** installed along the fluid passage **15**, such as within the second cavity **64**, and being adapted to block fluid flow between the first and second cavities **62**, **64**. The valve body **70** is adapted to move between a closed position where the valve body **70** obstructs the fluid passage **15** between the first and second cavities **62**, **64**, and an open position, where the valve body **70** is displaced such that fluid flow is enabled. As seen in FIG. 12, the internal surfaces of the body portion **16** can include abutment surfaces **17** shaped and sized to have the valve body **70** abut thereon for blocking fluid flow through the fluid passage **15**. The valve assembly **30** can also include a second resilient element **72**, such as a second spring **73**, configured to bias the valve body **70** in the closed position (e.g., bias the valve body **70** against the abutment surfaces **17**).

In addition, the valve assembly **30** can include a valve body actuator **71** configured to displace the valve body **70** from the closed position to the open position, thereby compressing the second spring **73** and enabling fluid flow. In this embodiment, the valve body actuator **71** includes a shaft **74** connected to and extending from the valve head **32** within the fluid passage **15**. It is thus noted that engaging the male component with the female component of the first fluid line causes the valve head **32** to retreat within the fluid passage **15**, and subsequently causes the shaft **74** to engage and push against the valve body **70** to move it in the open position. Therefore, the shank portion **18** can at least partially engage the female component prior to fluid communication being established between the first and second fluid lines due to the distance between the shaft head **77** and the valve body **70**.

In some embodiments, it can be desirable to at least partially control the pressure within the coupling **10**, such as within the fluid channel **25** and the fluid passage **15** (e.g., within the first cavity **62** and/or the second cavity **64**) to prevent malfunctions or damage caused by pressure differentials or pressurization issues. For example, hydraulic fluid can be provided to the second cavity **64** via the second fluid line, with the valve body **70** blocking fluid flow into the first cavity **62**. Therefore, it is noted that the pressure within the second cavity **64** increases, while the pressure within the first cavity **62** remains low (e.g., null), making it difficult to

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move the valve body 70 in the open position to provide hydraulic fluid to the first fluid line. As such, in some embodiments, the valve body 70 can include a pressure eliminator 76 operable to eliminate, or at least reduce, the pressure within the first and second cavities 62, 64 prior to moving the valve body 70 in the open position.

In this embodiment, and still referring to FIG. 12, the valve body 70 has a bore 75 defined therethrough and communicating with the first cavity 62 at a first end thereof, and with the second cavity 64 at a second end thereof. The pressure eliminator 76 can be coupled to the valve body 70 within the bore 75 and is operable to selectively eliminate the pressure within the second cavity to facilitate opening the valve body 70 (e.g., moving to the open position). More specifically, the pressure eliminator 76 can include an obstructing component, such as a plug 78, provided within the bore 75 and being shaped and configured to block fluid flow therethrough. The plug 78 can be displaced to enable a restricted flow of fluid from the second cavity 64 to the first cavity 62, thereby reducing (e.g., eliminating) the pressure within the second cavity 64, or at least creating a pressure balance between the first and second cavities (e.g., substantially the same pressure in each cavity). Therefore, displacement of the valve body 70 into the second cavity 64 to operate it in the open position is facilitated.

In some embodiments, the bore 75 can include a seat 80 adapted to have the plug 78 sit thereon when obstructing fluid flow through the bore 75. As seen in FIG. 12, in this embodiment, the seat 80 is provided proximate the first end of the bore 75 (i.e., proximate the first cavity 62), and the plug 78 engages the seat 80 to block fluid flow through the bore 75. It is noted that the plug 78 is movable between a seated position, where the plug 78 engages the seat 80 to block fluid flow, and an unseated position, where the plug 78 is displaced and spaced from the seat 80, thereby enabling fluid flow between the first and second cavities 62, 64. In this embodiment, the pressure eliminator 76 further includes a plug biasing element 82 operatively connected to the plug 78 and being configured to bias the plug 78 in the seated position. As will be described further below, the pressure eliminator 76 can be operated to apply a force on the plug 78 in order to displace and unseat it. Once the force is removed, the plug biasing element 82 is configured to revert the plug 78 in the seated position, thereby blocking fluid flow through the bore 75 once more.

In this embodiment, the pressure eliminator 76 further includes a cap 84 shaped and sized to fit within the bore 75 to at least partially restrict fluid flow therethrough. In this embodiment, the cap 84 is connected to the valve body 70 within the bore 75 via interference fit, although other connection methods are possible, such as via fasteners, via a keyjoint connection, via adhesive, etc. The cap 84 is illustratively provided proximate the second end of the bore 75, and is thereby on the side of the second cavity 64. As seen in FIG. 12, the plug biasing element 82 is connected to the cap 84 and extends between the cap 84 and the plug 78. Furthermore, the cap 84 includes a cap passage 85 defined therethrough for enabling fluid flow into the bore 75 from the second cavity 64. Therefore, when fluid is introduced within the second cavity 64 (e.g., via the second fluid line), the fluid can flow into the bore 75 via the cap passage 85, although fluid flow into the first cavity 62 is blocked by the plug 78 when in the seated position.

As previously described, engaging the coupling 10 with the female component of the first fluid line causes the valve head 32 to retreat within the fluid passage 15, and subsequently causes the shaft head 77 to engage and push against

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the valve body 70. Moreover, when in the seated position, the plug 78 can have a portion thereof extend into the first cavity 62. For example, in the illustrated embodiment of FIG. 12, the plug 78 includes a generally spherical body 79 configured to engage the seat 80 in a manner such that a portion of the spherical body 79 communicates with the first cavity 62. As such, moving the shaft 74 toward the valve body 70 causes the shaft head 77 to engage and push the plug 78 inwardly within the bore 75 (e.g., toward the second cavity 64) prior to engaging the valve body 70. It is thus noted that a restricted flowrate of fluid is enabled between the first and second cavities 62, 64 through the bore 75 due to the unseating of the plug 78 via the shaft head 77. It is further noted that enabling the fluid flow between the cavities 62, 64 causes the pressure within the second cavity 64 to drop, thereby facilitating displacement of the valve body 70 into the second cavity 64, and into the open position, therefore establishing a greater flowrate of fluid into the first cavity, and thus between the first and second fluid lines.

It is appreciated that the coupling 10 includes seals 90, or sealing elements, provided between various components for preventing fluid flow through the coupling 10 when fluid flow is undesired (e.g., when the coupling 10 is connected to only one of the first and second fluid lines). For example, the coupling 10 can include a shank portion seal 91 provided proximate the male component port 22 such that the shank portion seal 91 engages the valve head 32 prior to engaging the coupling 10 with the first fluid line, and as seen in FIG. 12. In addition, the coupling 10 can include an internal seal 92 provided about the valve body 70 such that when the valve body 70 is in the closed position, the internal seal 92 engages the abutment surfaces 17 to prevent fluid flow through interstices between the valve body 70 and the body portion 16. It is appreciated that the coupling 10 can include any suitable number of seals provided between any abutting components thereof.

The above-described implementations of the coupling provide for a quick-connect coupling, or a cartridge therefor (seen in FIG. 17), configured to connect to a complementary portion (e.g., male and/or female portion) and define a static connection therebetween. Particularly, the coupling 10 is adapted to block axial movement between the male and female portions, via the engagement of the ball bearings of the female portion with the groove of the male portion, for example. In addition, the restraining mechanism 40 is configured to block a rotational movement of the male and female components relative to one another. As such, it is appreciated that the coupling is adapted to block both axial and rotational movement between the male and female components, thereby defining the static connection therebetween. It should be noted that providing a static connection can increase lifespan of the coupling by reducing the stress, friction among other forces applied to various components of the coupling during operation.

Now referring to FIGS. 13 to 16, another implementation of the coupling is illustrated. In this embodiment, the coupling can be a swivelling coupling 100, where the male component 14 is rotatably coupled to the coupling adapter 24 such that the first fluid line can rotate relative to the second fluid line. In other words, the coupling 100 provides a swivelling connection between the first and second fluid lines. In this embodiment, the male component 14 includes a flange portion 102 extending radially outward from the body portion 16, at an end thereof opposite the shank portion 18. Furthermore, in this embodiment, the coupling adapter 24 includes a housing 104 defining an adapter cavity 105

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shaped and adapted to receive the body portion 16 of the male component 14 therein. As will be further described below, the swivelling coupling 100 also includes a nut 110 insertable within the housing 105 for at least partially securing the male component 14 therein.

In some embodiments, the adapter cavity 105 includes internal surfaces including at least one radial surface 106 and at least one axial surface 108. It should be understood that, as used herein, the expression "radial surface(s)" can refer to surfaces extending in generally perpendicular planes relative to the longitudinal axis (A) of the of the coupling 100. Similarly, it should be understood that, as used herein, the expression "axial surface(s)" can refer to surfaces being generally parallel to the longitudinal axis (A), and therefore perpendicular relative to the radial surfaces. The internal surfaces 106, 108 define the adapter cavity 105 which has an open end 107 through which the male component 14 is insertable. However, it is appreciated that other configurations are possible, for example, the housing 104 may include a plurality of cavities in which a corresponding number of male components and nuts may be introduced. The internal surfaces 106, 108 are preferably integrally formed with each other. This usually results from the machining of the housing 104 from a solid piece. In addition, the housing 104 includes an external surface 112 having any suitable finish and/or form. For example, the external surface 112 can have a generally cylindrical or circular shape. In the illustrated embodiment of FIGS. 13 to 16, the external surface 112 is curved and adapted to be retained within a retaining ring (not shown) mounted on the hydraulic machine and enable rotational freedom relative to the retaining ring.

In some embodiments, inserting the body portion 16 within the open end 107 of the adapter cavity 105 includes having the flange portion 102 cooperatively abut on the internal surfaces of the adapter cavity 105. Still referring to FIG. 16, it is appreciated that engaging the male component 14 within the housing 104 can establish fluid communication between the fluid passage 15 of the male component 14 and the fluid channel 25 of the coupling adapter 24. In this embodiment, the fluid passage 15 and fluid channel 25 are substantially collinear, although it is appreciated that other configurations are possible and may be used. For example, the fluid passage 15 and fluid channel 25 may have a variety of orientations, depending on the desired application, such as 90° angles or oblique angles.

In this embodiment, the flange portion 102 is integrally formed with the body portion 16 and extends therefrom in a manner such that the male component 14 has a generally T-shape. The flange portion 102 can be radially continuous and symmetrical, and can be disk-shaped and extend perpendicularly with respect to the body portion 16. As will be described below, this configuration of the male component 14 can provide support surfaces for abutting on the internal surfaces of the housing 104 and the nut 110, thereby distributing the forces (i.e. decreasing the pressure) applied on the components of the coupling 100. However, it is appreciated that the flange portion 102, and/or any other component of the male component 14, can have any suitable shape which cooperates with the internal surfaces of the housing 104. In some embodiments, the flange portion 102 can include scores (not shown) defined on an outer circumferential surface thereof for enabling fluid to flow between the flange portion 102 and the housing 104. This can be useful for lubricating purposes and for defining a self-lubricating coupling, such as the one described in Applicant's U.S. Pat. No. 8,047,579, which is incorporated herein by reference.

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Still referring to FIGS. 13 to 16, the nut 110 is securable within the adapter cavity 105 of the housing 104 and surrounding the male component 14 to radially restrain the same within the adapter cavity 105. In some embodiments, the nut 110 can be adapted to surround the male component 14 from the port 22 to the flange portion 102, which allows for improved support, stability and resistance to forces. In this embodiment, the nut 110 is secured about the body portion 16 of the male component 14, with the shank portion 18, and thus the port 22, extending beyond the nut 110 to facilitate connection with the first fluid line. In this embodiment, the nut 110 includes a projection 114 extending axially and internally into the cavity 105 of the housing 104 for axially restraining the flange portion 102, while allowing the male component 14 to rotate with respect to the housing 104 and the nut 110. The rotation of the male component 14 occurs about a longitudinal axis (A) of the shank portion 18.

In some embodiments, the nut 110 can have external threads and the internal surfaces of the housing 104 (i.e., the surfaces of the adapter cavity 105) can have corresponding internal threads to secure the nut 110 within the cavity of the housing. Alternatively, these components may be unthreaded and be bolted, clamped or otherwise connected to one another. As seen in FIGS. 13 to 16, the nut 110 includes a lip 116 extending over a perimeter rim 109 of the open end of the adapter cavity 105 to further secure the nut 110 in engagement with the coupling adapter 24.

In some embodiments, the hydraulic pressure within the adapter cavity 105 pushes the male component 14 axially toward the nut 110. Once pressurized, the pressure inside the housing 104 is substantially uniform in all directions and pushes perpendicularly on the surfaces (e.g., against the flange portion 102, and the radial and axial surfaces 106, 108). In operation, the fluid contained in the fluid passage 15 and/or fluid channel 25 is under hydraulic pressure. Operating pressures vary depending on the application, be it heavy or light. Typical ranges of hydraulic pressure in the forestry industry, for example, are between about 50 and about 4000 psi, and up to around 5000 psi in some cases. In load-sensing hydraulic circuits, the operating pressures most often vary between about 250 psi and 3000 to 4000 psi.

In some embodiments, this pressure results in the direct abutment of the flange portion 102 against the projection 114 of the nut 110. However, in this embodiment, the swivelling coupling 100 further includes a slide-ring 120 surrounding the body portion 16 and disposed between the nut 110 and the flange portion 102. The slide-ring 120 can be adapted to reduce the friction coefficient between the components, e.g., between the flange portion 102 and the nut 110, during rotation of the male component 14. The slide-ring 120 can be particularly desirable in high-pressure hydraulic systems (or alternatively in applications where the fluid is less lubricating) since the axial pressure on the male component 14 greatly increases the friction between the flange portion 102 and the nut 110. In some embodiments, the slide-ring 120 is composed of Nylatron® material group, Nyloil™, Nycast™, Teflon, or another suitable material for such a component. In some embodiments, the slide-ring 120 can include a thrust washer adapted to support the axial load applied thereto, such as the load applied thereto by the flange portion 102 during operation and/or during hydraulic shocks.

Referring to FIG. 16, when the male component 14 is coupled within the adapter cavity 105, the flange portion 102 defines an axial play 95 with the radial surface 106 of the adapter cavity 105. More particularly, the internal depth of the adapter cavity 105 is greater than the sum of the width

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of the flange portion 102 and the length of the projection 114 of the nut 110, thus enabling an amount of axial play 95. In some embodiments, the amount of axial play can be between about 0.005 and about 0.08 inches, although other configurations are possible, such as having a narrower axial play (e.g., <0.005 inches) or wider axial play (e.g., >0.08 inches).

In this embodiment, when under internal fluid pressure, the flange portion 102 is pushed toward the projection 114 of the nut 110, and thus abuts on the slide-ring 120. The slide-ring 120 is adapted to enable a distribution of forces, thereby decreasing the pressure between the components. This in turn enables the swivelling coupling 100 to have improved swivelling performance at higher pressures, such as in the range of 3000 to 5000 psi, for example. In some embodiments, the slide-ring 120 has a flat disk shape, but may also have an O-ring shape to reduce the coefficient of friction. It should be noted that, when the flange portion 102 is pressed against the slide-ring 120, the flange portion 102 is held in spaced relation to the internal axial surface 108 of the adapter cavity 105, and corresponds to the amount of axial play 95. It should be noted that the axial play 95 defined within the adapter cavity 105 may help protect the various components of the coupling 100 from hydraulic shocks (also called "hydraulic rams") or other types of fluid shocks or pressure differentials in and around the coupling.

In some embodiments, it can be desirable to maintain the contact between the flange portion 102, the slide-ring 120 and the nut projection 114 to increase sealing efficiency of the coupling 100 (e.g., prevent exterior fluid(s), dust and/or debris from entering the coupling). During operation, hydraulic fluid flowing through the coupling 100 urges the flange portion 102 against the slide-ring 120, thereby creating the contact between flange portion 102, slide-ring 120 and nut projection 114, and increases sealing efficiency. However, in some embodiments, the contact between these components can be broken, thereby compromising the seal integrity of the coupling. For example, and with reference to FIGS. 31 to 33, underwater applications without any prior pressurization of the coupling (FIGS. 31 and 32), vertical installations of the coupling where the flange portion 102 tends to move away from the slide-ring 120 under gravity (FIG. 33), or under the effect of hydraulic shocks and a vacuum/suction effect occurs within the coupling can decrease the seal integrity of the coupling by applying pressure on inner components of the coupling and breaking the contact between the flange portion 102 and the slide-ring 120, for example. Under such conditions, the risk of infiltration of dust, debris or other contaminants within the coupling is increased and can cause some issues.

Still with reference to FIG. 16, the swivelling coupling 100 can include a seal assembly 130 comprising at least one sealing ring 132 (or O-ring) adapted to cooperate with one or more of the male component 14, the nut 110 and the coupling adapter 24. In this embodiment, once the swivelling coupling 100 is assembled and in operation, the seal assembly 130 is pressed in between the male component 14 and the nut 110 to at least partially cut the pressure of any fluid leaking through the interstices of the coupling 100. The seal assembly 130 can be adapted to prevent fluids from leaking out of the coupling 100. In some embodiments, the seal assembly 130 may include O-rings, back-up rings, etc. Also, the seal assembly 130 may be easily replaced, in the event that it loses its efficiency. Various sealing joints may be used to cut the velocity of the fluid so that if the fluid leaks past one of the seals, it is less likely to leak past the next seal. In other words, a series of seals may be used as part of the seal assembly 130.

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In some embodiments, the seal assembly 130 further includes one or more outer seals 134 adapted to prevent fluid from leaking to an exterior of the coupling 100, or to prevent debris and dust from entering the coupling 100. With reference to FIGS. 13 to 16, in this embodiment, the outer seal 134 includes a wiper ring 136 surrounding the male component 14 proximate the shank portion 18. The wiper ring 136 is also adapted to engage the nut 110 in order to create a seal between the nut 110 and the male component 14. In some embodiments, the wiper ring 136 can have a flat-disk shape, although other configurations are possible, as will be described below.

In this embodiment, the male component 14 includes an outer groove 140 extending about the shank portion 18. The outer groove 140 can be shaped and adapted to receive a portion of the wiper ring 136 therein. As seen in FIGS. 14 and 16, the wiper ring 136 can include a ring portion 138 adapted to engage the nut 110, and a protruding portion 139 extending from the ring portion 138 and engaging the outer groove 140. It is noted that both the ring portion 138 and the protruding portion 139 surround the male component 14, i.e., extend circumferentially around the male component. In this embodiment, the outer groove 140 includes a groove wall 142 extending relatively perpendicularly relative to the shank portion 18, and thus substantially parallel to the front surface of the nut 110. The wiper ring 136 is therefore shaped and adapted to extend between the nut 110 and the groove wall 142, with the ring portion 138 engaging the nut 110, and the protruding portion 139 engaging the groove wall 142.

In some embodiments, the wiper ring 136 is made of a resilient material, such as rubber, and is adapted to protect the various components of the coupling during operation of the hydraulic lines (e.g., while fluid is flowing between the first and second fluid lines via the coupling). During operation, it is noted that the various components of the coupling 100 can be submitted to hydraulic shocks (also called "hydraulic rams") which can cause shifts, friction, damage, or failure of these components.

Some hydraulic shocks can result from a suction effect created by the pressure differential between fluid channel 25 and the fluid passage. Specifically, hydraulic fluid can be provided to the fluid channel 25 via the second fluid line, with the valve body 70 blocking fluid flow into the fluid passage 15. Therefore, it is noted that the pressure along the fluid channel 25 and within the adapter cavity 105 increases, while the pressure along the fluid passage 15 remains low (e.g., null). As such, when the valve body 70 is displaced in the open position, the low pressure within the fluid passage can create a vacuum causing the pressure within the adapter cavity 105 to drop substantially rapidly to match the pressure along the fluid passage 15 (e.g., 0 psi), thereby creating the suction effect within the coupling 100.

The suction effect (i.e., the hydraulic shock) often creates a force against the flange portion 102 which pushes the flange portion 102 against the nut 110. In this embodiment, the nut 110 then pushes against the wiper ring 136 provided about the male component. The resilient material of the wiper ring 136 is configured to be deformed, thus absorbing at least some of the forces created by the hydraulic shocks and reducing movement of the male component 14, slide-ring 120 and/or nut 110. Once the pressure stabilizes and/or is relieved, i.e., once the forces are no longer applied on the components of the coupling, the wiper ring 136 is adapted to revert back to its initial shape and configuration, thereby also moving the nut 110 and/or male component 14 back in position.

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As such, it is appreciated that each time the coupling 100 is connected and/or disconnected from the first fluid line, the pressure within the adapted cavity 105 increases and decreases accordingly, thereby creating repetitive suction effects. The wiper ring 136 is configured to at least partially counteract this suction effect to protect the components of the coupling 100 (e.g., the male component 14) and of the component(s) connected thereto (e.g., the female component), thereby increasing their lifespan and efficiency.

In some embodiments, the wiper ring 136 can have a tapered shape, where the diameter of the wiper ring at a first end thereof is greater than at a second and opposite end thereof. For example, in this embodiment, the protruding portion 139 extends at an angle from the ring portion 138 such that the inner diameter of the wiper ring 136 along the ring portion 138 is greater than the inner diameter of the wiper ring along the protruding portion 139. Moreover, the thickness of the wiper ring 136 along the protruding portion 139 is illustratively less than the thickness thereof along the ring portion 138. This configuration can improve the shock-absorption capabilities of the wiper ring 136 and also improve its resiliency (i.e., its ability to revert back to its initial shape and configuration). During operation, hydraulic pressure within the coupling can cause the wiper ring 136 to be squeezed between the nut 110 and the groove wall 142, thereby causing the protruding portion 139 to at least partially deform (e.g., buckle) to absorb at least some of the forces from the hydraulic rams, for example. The protruding portion 139 is adapted to revert back to its initial configuration once pressure within the coupling has stabilized.

In some embodiments, the ring portion 138 and protruding portion 139 can be made of the same material, and thereby have substantially the same resiliency and/or flexibility. However, it is appreciated that other configurations are possible. For example, the protruding portion 139 can be made of a more resilient or flexible material than the ring portion 138. In such embodiments, it is noted that the ring portion 138 can absorb some of the forces from the hydraulic shocks, although the more resilient protruding portion 139 is configured to absorb more forces.

As seen in FIG. 16, the second spring 72 illustratively extends between an internal surface of the adapter cavity 105 and the flange portion 102, and is adapted to also absorb some of the forces of the hydraulic shocks and bias the flange portion 102 against the slide-ring 120. Therefore, the components of the coupling (e.g., the flange portion 102, the slide-ring 120 and the nut 110) are adapted to remain in contact with one another to ensure sealing efficiency of the coupling. The second spring 72 can be optional, with the axial movement of the components of the coupling being at least partially blocked by the cooperation of the wiper ring 136 with the male component 14 (e.g., with the shank portion) and the nut 110. More specifically, under certain conditions, the flange portion 102 can be made to move into the axial play 95, thereby increasing the risks of impacts with the housing and breaking contact with the slide-ring 120. In this embodiment, when the male component 14 is pushed toward the axial play 95 (e.g., caused by hydraulic shocks, a vertical installation or under external pressure), the groove wall 42 engages the wiper ring 136. The wiper ring 136 can be adapted to absorb some of the forces, and also engages the nut 110, which in turn abuts against the housing 104 via the lip 116. As such, axial movement of the male component 14 toward the axial play 95 is blocked, or at least reduced.

It should be noted that reducing or blocking movement of the components of the coupling (e.g., the flange portion 102,

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the slide-ring 120 and the nut 110) can reduce the stress, deformation and impacts they sustain during operation. Reducing the forces applied on these components can increase their lifespan, which reduces the need to replace and/or repair them, thereby increasing the efficiency and lifespan of the coupling 100.

Referring broadly to FIGS. 1 to 16, it should be noted that, when the male component 14 of the coupling (10 or 100) engages the female component of the first fluid line, the restraining mechanism 40 is adapted to prevent axial movement of the male component 14 relative to the female component (e.g., via engagement of the ball bearing assembly within the groove 42 provided about the shank portion 18). Furthermore, the restraining mechanism 40 is also adapted to prevent rotational movement of the male component 14 about the longitudinal axis of the shank portion 18 relative to the female component via engagement of the detent 44 with the ball bearing assembly.

In addition, the flange portion 102 is shaped and sized to engage the adapter cavity 105, with the nut 110 blocking axial movement of the male component 14 toward the open end of the adapter cavity 105, and in some embodiments, the wiper ring 136 is adapted to block, or at least impede axial movement of the male component 14 toward the coupling adapter 24 (i.e., into the axial play 95). The axial movement of the male component due to the deformation and wear of the slide-ring 120 (e.g., the thrust washer), along with the suction effect and/or atmospheric pressure (e.g., when submerged in water) is thus also at least partially managed via the wiper ring 136, and the contact between the flange portion 102, slide-ring 120 and nut projection 114 is maintained. It should also be noted that greater forces applied on the wiper ring 136 increase the sealing efficiency of the wiper ring 136 since the wiper ring 136 is squeezed in place between the groove wall 142 and the nut projection 114, thereby further sealing interstices between the nut 110 and the male component 14. This can be desirable in certain situations, such as in vertical installations, or for underwater applications, where the coupling would not be required to be pressurized prior to being submerged since the increasing pressure from the water can increase the sealing efficiency of the wiper ring 136.

Now referring to FIGS. 17 to 20A, another implementation of the coupling 100 is shown. The male component 14 is provided within the housing 104 of the coupling adapter 24, with the shank portion 18 extending out of the housing 104 for connecting with a female component. The coupling 100 also includes the nut 110 for connecting the male component 14 to the coupling adapter housing 104. Referring more specifically to FIG. 20, the male component 14 can include a plurality of parts coupled to one another. For example, in this embodiment, the male component 14 comprises an inner part 180 which includes the flange portion 102 provided in the adapter cavity 105 proximate the coupling adapter 24, and an inner shaft 184 extending from the flange portion 102 toward the open end 107. Additionally, the male component 14 includes an outer part 182 which includes the shank portion 18. In this embodiment, the shank portion 18 is removably coupled to the inner part 180 and extends out the open end 107 of the adapter cavity 105. The shank portion 18 can include an outer shaft 186 shaped and adapted to connect with the inner shaft 184, thereby connecting the inner and outer parts of the male component 14 together.

In this embodiment, the inner and outer parts 180, 182 are connected together via an interference fit between the inner and outer shafts 184, 186, although other connection meth-

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ods are possible, such as with the use of fasteners, via a keyjoint connection, via adhesive, etc. The inner part **180** is adapted to extend within the outer part **182** (i.e., the inner shaft **184** extends within the outer shaft **186**). However, it is appreciated that other configurations are possible, such as having the outer part **182** extend within the inner part **180** in order to connect the two parts to one another. In the illustrated embodiment, the outer shaft **186** is further adapted to engage the nut **110** such that the outer shaft **186** is coupled between the inner shaft **184** and the nut projection **114**, with the nut projection **114** being coupled between the outer shaft **186** and the housing **104**. As such, it is noted that the inner and outer parts **180**, **182** of the male component **14** are secured to the housing via their cooperation with each other and with the nut **110**.

Still referring to FIG. 20, the shank portion **18** can include a cylindrical body **188** provided between the outer shaft **186** and the port **22**. The cylindrical body **188** is configured to engage the nut **110** to further connect the outer part **182** (e.g., the shank portion **18**) to the nut **110**, and thus to the housing **104**. The cylindrical body **188** generally has a greater diameter than the outer shaft **186** and the port **22**. The nut **110** can thus have a complementary shape to the outer part **182** of the male component **14**, for example, with the projection **114** of the nut **110** having an inner diameter adapted to receive the outer shaft **186**, but prevent insertion of the cylindrical body **188** therein. The nut **110** illustratively includes an outer end **111** having a greater diameter than the projection **114** and adapted to receive therein the cylindrical body **188** of the shank portion **18**.

In this embodiment, the outer end **111** defines a nut radial surface **113** adapted to face the cylindrical body **188**. As will be described further below, the male component **14** is adapted to engage the nut **110** and cooperate therewith to define the axial play **95** between the flange portion **102** and the coupling adapter housing **104**. Furthermore, the male component **14** engages the nut **110** in a manner such that the cylindrical body **188** is spaced from the nut radial surface **113**, defining therebetween a second axial play **195**. The axial play **95** and the second axial play **195** can generally have the same dimensions, although it is appreciated that one can be greater than the other, for example.

In some embodiments, the seal assembly includes various sealing elements, such as O-rings provided between the male component **14** and the nut **110**, between the nut **110** and the housing **104**, or a combination thereof. In this embodiment, and with reference to FIGS. 20 and 20A, the coupling **100** includes a plurality of sealing rings **132**, including a pair of sealing rings **132** provided within the outer end **111** of the nut **110**, between the nut **110** and the cylindrical body **188** of the shank portion **18**. The sealing rings **132** are configured to seal interstices between the nut **110** and the male component **14** (e.g., the cylindrical body **188**), and radially restrain the male component **14** within the nut **110**. Referring more specifically to FIG. 20A, the sealing rings **132** include a first sealing ring **132a** and a second sealing ring **132b** provided in a circumferential pocket **189** defined about the cylindrical body **188**. As mentioned, each sealing ring **132** extends between and engages the nut **110** (on an inner surface thereof) and the cylindrical body **188** (on an outer surface thereof), thereby radially restraining the cylindrical body **188** relative to the nut **110**.

The first sealing ring **132a** engages substantially flat surfaces, such that it radially restrains the male component **14** within the nut **110**, along with defining a seal therebetween. In this embodiment, the outer end **111** of the nut can have a tapered inner edge **115**, with the second sealing ring

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132b engaging the tapered inner edge **115**. As such, the second sealing ring **132b** can be adapted to radially restrain the male component **14** (e.g., the outer part **182**), and at least partially provide axial restraint to the male component as well. For example, and as described above, during operation, the coupling **100** can be subject to a suction effect which can pull the male component **14** further into the adapter cavity **105**, thus pulling the cylindrical body **188** further into the outer end **111** of the nut **110**. The tapered outer edge **115** can be shaped and configured to provide an abutment for the second sealing ring **132b**, which can then block axial movement of the cylindrical body **188** toward the nut **110**. In other words, the second sealing ring **132b** is adapted to be squeezed between the lateral surface of the circumferential pocket **189** and the tapered inner edge **115**, thereby preventing axial movement of the cylindrical body **188**. It should be noted that, by blocking axial movement of the cylindrical body **188**, the outer part **182** does not push against the inner part **180**, thereby maintaining the flange portion **102** in contact with the slide-ring **120** within the adapter cavity **105**. In some embodiments, the second sealing ring **132b** can be a pressure-energized seal, at least partially blocking both axial and radial movement of the male component, and enabling greater sealing efficiency as the pressure thereon increases.

It is appreciated that adjusting the angle at which the tapered inner edge **115** extends can correspondingly adjust the amount of axial restraint the second sealing ring **132b** can provide. For example, a tapered edge of about 45 degrees can be adapted to prevent axial movement over a greater range of pressures than a tapered edge of about 10 degrees. In some embodiments, the tapered inner edge **115** of the nut **110** can be angled relative to a nut axial surface **117** by about 5 to about 85 degrees, although other configurations are possible. It should also be noted that greater forces applied on the second sealing ring **132b** increase the sealing efficiency of the sealing ring **132b** since the cylindrical body **188** presses and squeezes the sealing ring **132** between the outer end of the nut **111**.

This can be desirable in certain situations, such as in vertical installations (seen in FIG. 33), or for underwater applications (seen in FIGS. 31 and 32), where the coupling is not required to be pressurized prior to being submerged since the increasing pressure from the water increases the sealing efficiency of the second sealing ring **132b**. For example, and as seen in FIGS. 31 and 32, one or more couplings **100** can be connected to an underwater pump, such as a pump configured to collect fish as part of a commercial fishing vessel, prior to being submerged. The pump can then be submerged without pre-pressurizing the coupling to prevent water from infiltrating the coupling. As seen in FIG. 32, the pump can be lowered to about 20 m under the surface, prior to providing fluid through the coupling, and the sealing ring **132** (i.e., the second sealing ring **132b**) is configured to prevent water and debris from entering the coupling **100**. It should be understood that increasing the sealing efficiency of the coupling when underwater is desirable to prevent water from infiltrating the coupling and mixing with the hydraulic fluid, which can cause issues such as freezing the fluids within the coupling, for example. It should be noted that the coupling connected to the pump can be a swivelling coupling **100**, or alternatively a non-swivelling coupling, with one or more swivelling couplings mounted along the pump conduits, such as at regular intervals (e.g., every 10 m).

In some embodiments, the outer part **182** of the male component **14** is removably connected to the inner part **180**

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and within the nut 110. As such, the outer part 182 can be disconnected from the inner part 180, thereby enabling interchangeability with another part. For example, the outer part 182 can be replaced with another male component outer part 182 (e.g., for maintenance or repair), or can be replaced with a corresponding female part. With reference to FIGS. 21 and 22, the coupling 100 can be adapted to define a female connection via the replacement of the outer part 182 of the male component, with a female component 190. The female component 190 connects to the inner part 180 in a similar fashion as the outer part 182 described above (e.g., with an outer shaft 186 coupled between the inner shaft 184 and the nut 110), and also engages the nut 110 via the first and second sealing rings 132a, 132b configured to at least partially block axial and radial movement of the female component relative to the nut 110 and housing 104.

It should be noted that the coupling can include any one of the above-described mechanisms adapted to prevent or block some form of movement of the components of the coupling. In some embodiments, the coupling can include both the restraining mechanism 40 and the outer seal 134 (e.g., the wiper ring 136, as seen in FIG. 13, or the tapered inner edge 115 and sealing rings 132, as seen in FIG. 20A), while in other embodiments, the coupling includes one of the restraining mechanism 40 and the outer seal 134, as seen in FIG. 1 (i.e., restraining mechanism 40 only), for example. It should also be noted that the coupling can be adapted to provide connection between a pair of fluid lines, as described herein, although other configurations are possible. For example, the coupling may also interconnect more than two fluid lines, where the male component 14 can include a plurality of ports 22 and can be connectable to a corresponding number of fluid lines. In some embodiments, the coupling can be coupled between an apparatus and a first hydraulic line, whereby the apparatus provides the flow of hydraulic fluid.

Referring to FIGS. 23 to 25C, the quick release male coupling and related components (e.g., the wiper ring 136 and/or the tapered inner edge 115 and sealing rings 132) and mechanisms (e.g., the restraining mechanism 40) described herein can be included in various types of couplings adapted for various applications. For example, the restraining mechanism 40 can be provided on a coupling cartridge 10a (FIG. 23) adapted for connection with a coupling housing, such as a swivelling coupling housing (FIG. 24) adapted for applications requiring manipulations of the hydraulic lines in a 3D space, such as in the forest industry. In other embodiments, the coupling can include a manifold-type coupling, which can be a quick release male coupling (FIG. 25). Alternatively, and with reference to FIGS. 25B and 25C, the coupling can correspond to a swivelling manifold-type coupling which can include quick-lock (or quick-release) capabilities 100b, threaded male connection(s) 100c, a female connection 100d and/or a «screw-to-connect coupling with swivel» 100e. However, it is appreciated that other configurations are possible, and that the couplings can have any suitable size and/or shape. In addition, and as seen in FIG. 26, the coupling can be part of a multiport assembly 150, whereby a plurality of couplings, such as swivelling couplings 100, can be coupled to a locking plate 152 which can be connected to hydraulic machinery in any suitable manner.

Now referring to FIGS. 27 to 30, the coupling 10 can be provided with a crimp fitting 160 attachable to one end of the coupling 10, such as to the male component 14. It should be noted that the crimp fitting, or crimped sleeve 160, is adapted to create custom hydraulic hose assemblies, and

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defines a crimped end 162 of the coupling 10 (i.e., the end provided with the crimp fitting 160) adapted to be connected to a hydraulic line (e.g., the first fluid line). The crimped end 162 is securable to the hydraulic line via a crimping machine or device, creating a permanent connection therebetween for repairing or creating hydraulic line assemblies. In this embodiment, the coupling can include a plurality of sealing rings 132, including an intermediary sealing ring 132c (e.g., an O-ring) between the crimped sleeve and the projection 114 of the nut 110 configured to block axial movement of the crimped sleeve 160 relative to the nut 110 and housing 104, thereby blocking axial movement of the male component 14 in a similar fashion.

It is appreciated that in the implementations described herein, the couplings can have increased sealing efficiency against external pressures, such as in underwater applications, for example. Moreover, the sealing efficiency of these couplings can increase as the surrounding pressure increases as the seals surrounding the coupling are squeezed into position, further sealing the interstices in the coupling. In some embodiments, the sealing elements can provide axial and radial contact with one or more component of the coupling. These sealing elements can be adapted to at least partially block radial and axial movement of the one or more components, thereby reducing the stress applied thereon and increasing their lifespan.

The present disclosure may be embodied in other specific forms. The described example implementations are to be considered in all respects as being only illustrative and not restrictive. For example, it should be understood that the shape of the male component 14 may be modified in a plurality of ways. Similarly, the internal shape of the adapter cavity 105 can be modified and should have a corresponding shape to house the flange portion 102, and to provide support surfaces for distributing forces while allowing adequate play for lubrication and movement. Furthermore, the flange portion 102 may have surfaces that are bevelled or curved. These bevelled or curved surfaces can affect the fluid movement between the surfaces of the coupling and also affect the force distribution. Since the pressurized fluid exerts pressure perpendicularly with respect to the solid surfaces of the coupling components, it is appreciated that varying the angles and curvatures and/or surface area of the male component, different force-distribution effects may be achieved.

In addition, one or more of the components of the coupling(s) and/or steps of the method(s) described herein could be modified, simplified, altered, omitted and/or interchanged, without departing from the scope of the present disclosure, depending on the particular applications which the coupling is intended for, and/or the desired end results, as briefly exemplified herein and as also apparent to a person skilled in the art. For example, the restraining mechanism 40 can be modified to enable interaction and cooperation with a different component than a ball bearing assembly. As seen in FIGS. 34 and 35, in some embodiments, the internal components of the female portion can include one or more latching elements 300 having an elongate shape, and which can extend at least partially circumferentially within the port of the female portion, for example. As such, the male component can include complementarily-shaped detents (e.g., recesses 52) for receiving therein the latching elements 300. It is noted that the latching elements 300 can be provided at regular intervals about the male component 14 (e.g., about the shank portion 18), such as at 90 degrees intervals, although other configurations are possible.

The present disclosure intends to cover and embrace all suitable changes in technology. The scope of the present disclosure is, therefore, described by the appended claims rather than by the foregoing description. The scope of the claims should not be limited by the implementations set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

As used herein, the terms “coupled”, “coupling”, “attached”, “connected” or variants thereof as used herein can have several different meanings depending in the context in which these terms are used. For example, the terms coupled, coupling, connected or attached can have a mechanical connotation. For example, as used herein, the terms coupled, coupling or attached can indicate that two elements or devices are directly connected to one another or connected to one another through one or more intermediate elements or devices via a mechanical element depending on the particular context.

In the above description, the same numerical references refer to similar elements. Furthermore, for the sake of simplicity and clarity, namely so as to not unduly burden the figures with several references numbers, not all figures contain references to all the components and features, and references to some components and features may be found in only one figure, and components and features of the present disclosure which are illustrated in other figures can be easily inferred therefrom. The implementations, geometrical configurations, materials mentioned and/or dimensions shown in the figures are optional, and are given for exemplification purposes only.

In addition, although the optional configurations as illustrated in the accompanying drawings comprises various components and although the optional configurations of the couplings as shown may consist of certain geometrical configurations as explained and illustrated herein, not all of these components and geometries are essential and thus should not be taken in their restrictive sense, i.e. should not be taken as to limit the scope of the present disclosure. It is to be understood that other suitable components and cooperations thereinbetween, as well as other suitable geometrical configurations may be used for the implementation and use of the couplings, and corresponding parts, as briefly explained and as can be easily inferred herefrom, without departing from the scope of the disclosure.

The invention claimed is:

1. A quick release coupling for providing a connection between a first fluid line provided with a female component comprising a ball bearing assembly and a second fluid line, the quick release coupling comprising:

a male component having a fluid passage therethrough and comprising:

a body portion having an internal surface at least partially defining the fluid passage; and

a shank portion extending from the body portion and being adapted to engage the female component of the first fluid line, the shank portion having a shank portion port adapted to establish fluid communication between the first fluid line and the fluid passage, the shank portion comprising a restraining mechanism adapted to engage the ball bearing assembly and block a rotational movement and an axial movement of the male component relative to the first fluid line when engaging the first fluid line; and

a coupling adapter having a fluid channel therethrough and being securable to the male component proximate the body portion to establish fluid communication between the fluid channel and the fluid passage, the

coupling adapter being adapted to engage the second fluid line and comprising an adapter port adapted to establish fluid communication between the second fluid line and the fluid channel, the coupling adapter further comprising a housing having internal surfaces comprising at least one radial surface and at least one axial surface, said internal surfaces defining a cavity having an open end, wherein the body portion of the male component is shaped and sized to engage the cavity via the open end and abut the at least one radial surface; and

a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the body portion within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivellably interconnected.

2. The quick release coupling according to claim 1, wherein the shank portion comprises an external surface, and wherein the restraining mechanism is provided along the external surface of the shank portion.

3. The quick release coupling according to claim 1, wherein the restraining mechanism comprises a groove extending circumferentially about the shank portion, the groove being shaped and adapted to receive the ball bearing assembly for blocking the axial movement of the male component.

4. The quick release coupling according to claim 3, wherein the groove extends circumferentially about the shank portion in a single plane.

5. The quick release coupling according to claim 3, wherein the groove extends partially about the shank portion such that the groove comprises a first end and a second end spaced from one another and defining an ungrooved portion therebetween.

6. The quick release coupling according to claim 5, wherein the ungrooved portion is shaped and adapted to engage the ball bearing assembly between a pair of adjacent bearing balls of the ball bearing assembly to block the rotational movement of the male component.

7. The quick release coupling according to claim 1, wherein the restraining mechanism further comprises a detent adapted to engage the ball bearing assembly to block the rotational movement of the male component.

8. The quick release coupling according to claim 7, wherein the detent comprises a plurality of recesses provided along a groove for receiving respective bearing balls of the ball bearing assembly to block the rotational movement of the male component.

9. The quick release coupling according to claim 7, wherein the detent comprises a protrusion extending from the shank portion, the protrusion being shaped and sized to engage the ball bearing assembly between a pair of adjacent bearing balls of the ball bearing assembly to block the rotational movement of the male component.

10. The quick release coupling according to claim 9, wherein the restraining mechanism comprises a groove extending circumferentially about the shank portion, and wherein the protrusion extends from within the groove.

11. The quick release coupling according to claim 9, wherein the protrusion is integrally formed with the shank portion or is welded thereto.

12. The quick release coupling according to claim 1, further comprising a valve assembly operable to control

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fluid flow along the fluid passage of the male component before, during and after engaging the shank portion with the first fluid line.

13. The quick release coupling according to claim 12, wherein the valve assembly comprises a valve head provided within the fluid passage proximate the shank portion port and being operable in a closed position where fluid flow is blocked through the shank portion port, and an open position, the valve assembly further comprising a head spring connected to the valve head and being adapted to bias the valve head in the closed position.

14. The quick release coupling according to claim 12, wherein the valve assembly comprises a valve body provided within the fluid passage along the body portion and being operable in a closed position where fluid flow through the fluid passage is blocked, and an open position where fluid flow through the fluid passage is allowed, the valve assembly further comprising a body spring connected to the valve body and being adapted to bias the valve body in the closed position.

15. The quick release coupling according to claim 1, further comprising a thrust washer surrounding the shank portion between the nut and the body portion.

16. The quick release coupling according to claim 1, wherein the shank portion comprises a flange extending radially outward and having a flange surface being substantially perpendicular to an external surface of the shank portion and facing the nut, and wherein the quick release coupling further comprises an outer sealing element surrounding the shank portion and extending between the flange surface and the nut.

17. A quick release coupling for providing a connection between a pair of hydraulic lines, the quick release coupling comprising:

- a male component having a fluid passage therethrough and comprising:
 - a shank portion adapted to engage a first hydraulic line, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line of the pair of hydraulic lines and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line; and
 - a coupling adapter connectable to the male component and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the

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coupling adapter being adapted to engage a second hydraulic line of the pair of hydraulic lines and comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel, the coupling adapter further comprising a housing having internal surfaces defining a cavity having an open end, wherein the male component is shaped and sized to engage the cavity via the open end and abut at least one of the internal surfaces; and

- a nut securable within the cavity of the housing and surrounding a section of the shank portion to radially restrain the male component, the nut being adapted to axially restrain the male component within the cavity and allow the male component to rotate with respect to the housing and the nut about a longitudinal axis of the shank portion, thereby enabling the first and second fluid lines to be swivellably interconnected.

18. A quick release coupling for providing a connection between a pair of hydraulic lines, the quick release coupling comprising:

- a male component having a fluid passage therethrough and comprising:
 - a shank portion adapted to engage a first hydraulic line of the pair of hydraulic lines, the shank portion having a first port adapted to establish fluid communication between the first hydraulic line and the fluid passage, the shank portion comprising a restraining mechanism adapted to cooperatively engage the first hydraulic line and block a rotational movement and an axial movement of the male component relative to the first hydraulic line when engaged with the first hydraulic line; and
 - a coupling adapter comprising a housing shaped and adapted to receive the male component therein and having a fluid channel therethrough adapted to be in fluid communication with the fluid passage, the coupling adapter being configured to axially restrain the male component while enabling rotational movement of the male component within a cavity and about a longitudinal axis of the shank portion,

the coupling adapter being adapted to engage a second hydraulic line of the pair of hydraulic lines such that the first and second hydraulic lines are swivellably connected, the coupling adapter further comprising a second port adapted to establish fluid communication between the second hydraulic line and the fluid channel.

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